

Utility-Preserving Federated Graph Learning With Dual-Perspective Fairness

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Abstract—Fairness-aware federated graph neural networks (FedGNNs) necessitate consideration of both the server and the clients. However, fairness-aware methods struggle to enhance dual-perspective (i.e., server and clients) fairness without sacrificing utility due to the distributed learning framework. As a consequence, the utility sacrifices of fairness-aware graph learning methods are even exacerbated in federated frameworks. In this work we propose F3GL, a dual-perspective fairness federated graph learning method that enhances both global (for the server) and local fairness (for clients) while preserving utility. Through theoretical analysis, we delineate the similarity between original sensitive features and those after convolution under different spectra. Our findings reveal that only the principal eigenvalue contributes to enhancing this similarity. Moreover, our theoretical analysis applies universally to both clients and servers. Specifically, employing a specialized eigenvalue selection strategy allows for effective optimization of both local and global fairness. Drawing on these insights, we improve dual-perspective fairness through the lens of spectral graph theory without sacrificing utility. Experimental results on two real-world datasets show the superiority of F3GL over existing baselines.

Index Terms—Graph neural network, federated learning, local fairness, global fairness, spectral graph learning.

I. INTRODUCTION

FEDERATED graph neural networks (FedGNNs), a trending distributed learning paradigm, effectively addresses the privacy concerns inherent in graph neural networks (GNNs) [1]. Despite their success, most FedGNNs inevitably overlook the bias in the graph data, resulting in discriminatory predictions towards sensitive subgroups, such as gender, age, nationality,

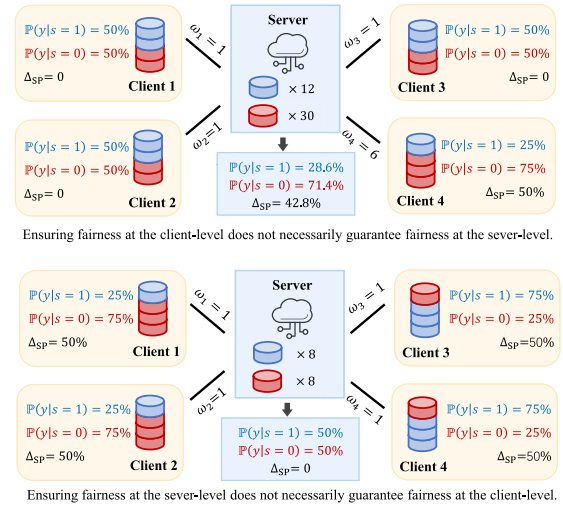


Fig. 1. A toy example of the disparity between local fairness and global fairness.

and race [2]. Algorithms lacking fairness considerations may cause social discrimination, increasing the risk of fraud for specific demographic groups and restricting their access to various services [3]. Moreover, in federated settings, the disparate distribution of data across clients exacerbates fairness concerns [4]. The deployment of FedGNNs in real-world scenarios has thus presented a significant challenge due to potential fairness issues. To harness the full potential of FedGNNs in critical applications, such as salary evaluation [5] and traffic management [6], it is imperative to ensure fair predictions for individuals with diverse sensitive features.

Fairness-aware GNNs aim to achieve model predictions that are independent of sensitive features, as evaluated by independence metrics like statistical parity and equality opportunity [7]. In FedGNNs, fairness needs to be evaluated from a dual-perspective: global and local. Local fairness focuses on the fairness of the model within each individual client before aggregation, capturing the potential bias introduced by non-IID sensitive attribute distributions across clients. Global fairness, on the other hand, measures fairness at the server level after aggregation, reflecting how the combined model behaves across the overall population. This dual-perspective evaluation is essential, as achieving fairness on either side alone does not guarantee fairness in the overall federated system. As shown in Fig. 1,

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TABLE I
THE SACRIFICING ACCURACY (%) OF EACH METHOD

Metrics	FairSIN	NIFTY	FairVGNN	FairGNN
$\Delta\text{ACC}_{\text{Vanilla}}$	-0.37	-0.90	-4.60	-2.20
$\Delta\text{ACC}_{\text{Local}}$	-2.39	-3.19	-6.50	-7.19
$\Delta\text{ACC}_{\text{Global}}$	-1.89	-3.02	-4.62	-7.42

ensuring fairness with one perspective does not necessarily guarantee fairness with another perspective. Wang et al. [8] further confirmed that global fairness cannot be expressed as a linear combination of local fairness.

The distributed nature of federated learning, which lacks comprehensive graph neighbour information, complicates the enhancement of both local and global fairness using message-passing fairness-aware methods [9]. As a result, current fairness-aware solutions for graph learning methods cannot be readily applied to tackle the specific fairness issues in FedGNNs. Moreover, these fairness-aware methods inevitably sacrifice algorithmic utility, often assessed in terms of prediction accuracy [10]. Within the distributed learning paradigm, the utility sacrificed by fairness-aware methods in FedGNNs is more significant. For example, we compare the utility sacrifices of four fairness-aware methods in both vanilla GNN and FedGNN frameworks on a real-world dataset (Pokey-z), as illustrated in Table I. Here, $\Delta\text{ACC}_{\text{Vanilla}}$ denote the utility sacrifice of vanilla methods, while the baseline is GCN. $\Delta\text{ACC}_{\text{Local}}$ and $\Delta\text{ACC}_{\text{Global}}$ denotes accuracy at client-level and server-level in FedGNNs framework, respectively. The results indicate that the utility sacrifice is even more significant in the FedGNN framework.

This paper focuses on the correlation between the graph spectrum of each client and dual-perspective fairness. Our analysis revealed the following insights: 1) The fairness at the client-level before aggregation and at the server-level after aggregation primarily depends on the eigenvector corresponding to the largest magnitude eigenvalue of the adjacency matrix. Selecting this eigenvector as the spectrum of convolution helps maintain fairness in prediction results. Spectral graph theory indicates that the largest nonzero eigenvalue is related to the geometric structure of the graph (specifically, algebraic connectivity and spectral radius), ensuring algorithmic utility. 2) In finite-layer convolutions, the convergence rate depends mainly on the first two largest eigenvalues and decreases exponentially. The impact of non-principal eigenvalues diminishes with more convolutional layers, but their influence on the independence of sensitive features increases, reducing algorithmic fairness. Based on these analyses, we propose that an eigenvector selection mechanism can guarantee dual-perspective fairness while preserving utility.

We present a novel approach, **F3GL** (global Fairness and local Fairness Federated Graph Learning). F3GL is designed to improve both local fairness and global fairness while maintaining algorithmic utility through an optimized spectrum. Firstly, F3GL employs the FedAvg algorithm as the federated learning framework, weighting the updated local averaging parameters based on the different sizes of the client-side data. Secondly, in each client, F3GL computes e largest magnitude eigenvalues along with their corresponding eigenvectors from the adjacency

matrix. This subset of eigenvalues captures the essential spectral properties related to sensitive features while mitigating the influence of non-principal eigenvectors. Thirdly, F3GL uses a learnable Fourier feature positional encoding function to further enhance the utility of the model by providing a richer representation of the graph's spectral properties. Finally, F3GL incorporates the optimized spectrum into the graph convolution process, achieving fair node representations and ensuring that the model maintains high utility. In summary, our contributions are outlined below:

- *A novel approach for addressing dual-perspective fairness issues in federated graph learning:* To enhance fairness in federated graph learning under dual-perspective, we present F3GL, a novel spectral graph learning method designed to consider fairness at both the client- and server-level. F3GL strategically mitigates the convolution impact on sensitive features both before and after client aggregation.
- *Theoretical insights into sensitive features expression:* By investigating the relationship between sensitive features and spectrum within FedGNNs in clients and the server, we unveil a mathematical framework that leverages the largest magnitude eigenvalues of the client-level adjacency matrix. This foundation enables a quantifiable understanding of the correlation between sensitive features and spectrum in a federated framework.
- *An eigenvalue selection mechanism for enhancing dual-perspective fairness while preserving utility:* Building on our theoretical findings, we introduce an innovative eigenvalue selection mechanism that truncates the spectrum to ensure dual-perspective fairness while preserving utility. This nuanced approach effectively addresses fairness challenges within FedGNNs.
- *Empirical validation through extensive experiments:* To verify the effectiveness of the proposed F3GL approach, we conduct comprehensive empirical evaluations on two real-world datasets with multi-sensitive features. The results demonstrate F3GL's superior performance in achieving local fairness, global fairness, and utility compared to state-of-the-art fairness-aware GNNs.

The rest of the paper is organized as follows. In Section II, we review the related work. Section III covers the essentials of FedGNNs, fairness evaluation metrics, federated graph learning fairness evaluation metrics, and some proposed lemma. In Section IV, we delve into the theoretical findings that underpin F3GL. Section V presents the development of F3GL. We report the experimental results in Section VI. Finally, we conclude the paper in Section VII.

II. RELATED WORK

A. Federated Graph Neural Networks

FedGNNs are designed for scenarios where explicit graph data exist but are isolated across clients. In such cases, GNNs require federated learning algorithms to assist in training models with this distributed data [11]. Current federated graph learning approaches can be broadly categorized into two main types of

frameworks: It integrates GNNs with different federated frameworks, focusing on using various federated learning frameworks to train GNN models, potentially leveraging the unique features of each framework to improve performance or address specific challenges in federated graph learning. Adapting GNN models to the underlying federated framework involves modifying existing GNN models to work within the constraints and requirements of federated learning.

FedGNN methods focus on optimizing the federated framework itself. For example, Lei et al. [12] propose FedCog, a novel federated graph learning framework that efficiently handles coupled graphs. These are a type of distributed graph data commonly found in real-world applications such as mobile carriers' communication networks and banks' transaction networks. FedGCN [13] utilizes federated learning to train Graph Convolutional Network (GCN) models for semi-supervised node classification, offering fast convergence and minimal communication overhead. FED-PUB [14] addresses personalized sub-graph federated learning by focusing on improving interrelated local GNNs rather than learning a single global model. F² GNN [15] enhances group fairness during the aggregation stage of the federated learning framework, though it incurs significant utility loss in basic FedGNN architectures.

Furthermore, previous studies have explored different GNNs within the basic federated framework. For example, Semi-GraphFL [16] targets graph classification tasks with partially labeled data, introducing an auxiliary loss for unlabeled data to guide the training process toward more effective predictions. GraphFL [17] addresses challenges in semi-supervised node classification on graphs, such as poor performance when data across clients are non-IID (independent and identically distributed), inability to handle data with new label domains, and inability to leverage unlabeled data, which are common in real-world graph-based problems. FedLit [18] dynamically detects latent link types during FedGNN training using an EM-based clustering algorithm, differentiating message-passing through different types of links using multiple convolution channels.

Despite these advancements, it is imperative to acknowledge that prevailing FedGNNs typically lack explicit considerations for algorithmic fairness. Results obtained from applying these methods to real-world datasets often expose discernible fairness issues, highlighting the need for approaches that explicitly address fairness concerns.

B. Fairness-Aware GNNs

Fairness in GNNs has garnered significant attention, particularly in identifying and mitigating discrimination related to specific sensitive features. Many fairness-aware GNN methods aim to preserve the independence of sensitive features through pre-processing and in-processing techniques [19].

Current fairness-aware GNNs improve fairness by pre-processing the dataset. FairVGNN [20] generates fair views of features by automatically identifying and masking sensitive-correlated features, taking into account correlation variation after feature propagation. It employs two feature masking strategies based on feature correlations, emphasizing the importance

of considering feature propagation and correlation variation in mitigating discrimination. Graphair [21] is designed to automatically identify fairness-aware augmentations from input graph data, aiming to circumvent sensitive features while preserving the integrity of other valuable data. FairAC [22] presents a fair feature completion approach to address information gaps and obtain equitable node embeddings for graphs lacking features. However, in federated settings, datasets are distributed across various local ends, and the lack of centralized data complicates the implementation of pre-processing methods.

Some in-processing methods focus on constraining the influence of algorithms on sensitive features. FairGNN [23] is designed to limit the impact of sensitive features using an estimation function and an adversarial debiasing loss function. FairSIN [24] introduces a novel neutralization-based paradigm, incorporating additional fairness-facilitating features into node features or representations before message-passing. FMP [25] explicitly addresses the use of sensitive features in forward propagation for node classification tasks using cross-entropy loss, eliminating the need for data pre-processing.

Despite these advancements, enhancing fairness comes at the cost of algorithmic utility. The distributed nature of federated learning exacerbates the utility sacrifice associated with fairness-aware algorithms. FUGNN [26] utilizes spectral graph methods to improve fairness and utility simultaneously based on theoretical analysis. However, it is limited to a trade-off between single fairness and utility as well as requires considering the full graph spectrum, which is challenging for individual clients in a federated framework. Consequently, FedGNNs achieve dual-perspective fairness without sacrificing utility remains a significant challenge.

C. Spectral GNNs

Spectral GNNs process features through filters, with well-known models including GCN [27], Simplified Graph Convolution (SGC) [28], and Simple Spectral Graph Convolution (S²GC) [29]. These models leverage eigenvectors of the (normalized) Laplacian for Graph Fourier Transform [30].

Polynomials are utilized to approximate spectral graph convolution and are optimized to enhance the utility of GNNs. For instance, Lu et al. [31] propose an eigenvalue correction strategy that liberates polynomial filters from the constraints of repeated eigenvalue inputs. This strategy enhances the uniform distribution of eigenvalues, thereby mitigating repeated eigenvalues and improving the fitting capacity and expressive power of polynomial filters. AFGNN [32] introduces an adaptive filtering GNN model capable of capturing all frequency information on large-scale graphs. It incorporates node features into several spectral graph learning tasks. Mao et al. [33] propose a non-IID PAC-Bayesian generalization bound for GNNs, shedding light on the reasons for performance disparities, namely the aggregated feature distance and homophily ratio difference between training and testing nodes. FE-GNN [34] conducts a comprehensive examination of the dominant feature space in representation learning, optimizing the utility of GNNs from the perspective of the dominant feature space. Amico et al. introduce a method

TABLE II
NOTATIONS

Notations	Definitions
$\mathcal{G}$	The undirected graph
$\mathcal{V}$	The node set
$\mathbf{X}$	The node feature matrix
$\mathbf{H}^{(l)}$	The output for the l -th graph convolution layer
$\mathbf{A}$	The adjacency matrix
y	The ground truth label
$\hat{y}$	The prediction label
s	The sensitive feature
γ_i	The eigenvalue of adjacency matrix
$\mathbf{p}_i$	The eigenvector of adjacency matrix
$\mathcal{C}_i$	The client in federated learning
$\mathcal{G}_i$	The unique private graph for client $\mathcal{C}_i$
$\mathbf{A}_i$	The adjacency matrix for client $\mathcal{C}_i$
$\mathcal{S}$	The server in federated learning

that emulates graph convolution via eigenvectors leveraging, bypassing the need for time-consuming gradient descent training procedure in recommender system.

Within conventional GNN architectures, the interdependence between different channels of the filter can hinder the utility of more potent filters. To address this concern, PDF [35] introduces a novel normalized adjacency matrix utility that can leverage an expanded set of bases and learnable filter coefficients, thereby enhancing the responsiveness of the filter. Specformer [36] further elaborates on this methodology, extending its applicability from tasks primarily focused on the graph level to encompass tasks at the node level.

However, these spectral filters do not explicitly consider algorithmic fairness. They overlook the interdependence between sensitive and other features, potentially introducing bias into the filter. This gap highlights the need for approaches that incorporate fairness considerations into spectral graph learning methods, ensuring that the models do not inadvertently propagate or amplify biases present in the data.

III. PRELIMINARIES

A. Notations

We denote sets with copperplate uppercase letters such as $\mathcal{A}$, matrices with bold uppercase letters such as $\mathbf{A}$, and vectors with bold low-case letters such as $\mathbf{a}$. We use rules similar to NumPy in Python for matrix and vector indexing. Given a matrix $\mathbf{A}$, $\mathbf{A}[i, j]$ denotes the element in the i -th row and the j -th column. $\mathbf{A}[i, :]$ and $\mathbf{A}[:, j]$ denote the i -th row and the j -th column, respectively.

We denote an undirected graph as $\mathcal{G} = (\mathcal{V}, \mathbf{A}, \mathbf{X})$, where $\mathcal{V}$ is the set of n nodes, $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the adjacency matrix, $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the node feature matrix, and d is the feature dimension. For the l -th graph convolution layer, we generalize spectral graph convolution as follows [37], [38]:

$$\mathbf{H}^{(l)} = (1 - \theta)\mathbf{A}\mathbf{H}^{(l-1)} + \theta\mathbf{H}^{(0)}, \quad (1)$$

where $\theta \in (0, 1)$, $\mathbf{H}^{(0)} = f_{\Theta}(\mathbf{X})$. $h \in \mathbb{R}^{1 \times n}$ is one channel in the filter that corresponds to one dimension of $\mathbf{H}^{(0)}$. We present a summary of commonly used notations in Table II.

In particular, we denote the sensitive feature channel as h_s . Based on the generalized formulation (1), we analyze fairness in existing graph convolution from the perspective of the graph's spectrum. Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric matrix with real-valued entries. The eigenvalues of $\mathbf{A}$ are ordered as $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$, with corresponding eigenvectors $\mathbf{p}_i$ ($i \in \{1, 2, \dots, n\}$). The sensitive features h_s after one layer convolution, is represented as $\mathbf{A}h_s$. After l layers convolution, it is represented as $\mathbf{A}^l h_s$. The cosine similarity between $\mathbf{A}^l h_s$ and h_s , denoted as $\cos(\langle \mathbf{A}^l h_s, h_s \rangle)$, measures the similarity between original sensitive features and the sensitive features after l layers convolution. A higher cosine similarity indicates a stronger preservation of sensitive feature independence, reflecting higher fairness in GNNs. Furthermore, we measure the eigenvector's influence on the sensitive features by $\sum_{i=1}^e \cos(\langle h_s, \mathbf{p}_i \rangle)$, where e is the number of selection eigenvectors.

B. Federated Graph Neural Network

This study explores a horizontal federated learning (HFL) setting, involving K clients $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$ and a server $\mathcal{S}$ [39]. Each client $\mathcal{C}_i$ possesses a unique private graph $\mathcal{G}_i$, with different node sets $\mathcal{V}_i$ while sharing the same set of node features. The adjacency matrix of each client is denoted as $\mathbf{A}_i \in \mathbb{R}^{|\mathcal{V}_i| \times |\mathcal{V}_i|}$. For a matrix $\mathbf{A}'_i \in \mathbb{R}^{n \times n}$ and nodes $v_p, v_q \in \mathcal{V}$, we define:

$$\mathbf{A}'_i[p, q] = \begin{cases} \mathbf{A}[p, q] & v_p, v_q \in \mathcal{V}_i \\ 0 & \text{else} \end{cases} \quad (2)$$

In HFL, clients maintain their local graphs without sharing them with the server $\mathcal{S}$, primarily for privacy reasons. The objective of HFL is to optimize the global objective function while preserving the privacy of local graphs:

$$\arg \min_{\omega} \sum_{i=1}^K \mathcal{L}_i(\omega) = \arg \min_{\omega} \sum_{i=1}^K \sum_{j=1}^{n_i} \ell_i(v_j, \mathcal{G}_i; \omega), \quad (3)$$

where $\mathcal{L}_i(\omega)$ is the loss function for client $\mathcal{C}_i$, parameterized by ω , n_i is the number of nodes in the local graph $\mathcal{G}_i$, and $\ell_i(v_j, \mathcal{G}_i; \omega)$ denotes the average loss over the local graph of client $\mathcal{C}_i$.

In this work, we adopt the federated averaging (FedAvg) framework to our context [40]. In FedAvg, only model parameters are exchanged between the server $\mathcal{S}$ and each client $\mathcal{C}_i$. Specifically, in each round t , the server $\mathcal{S}$ sends the current global model parameters ω_t to a selected subset of clients for local training. Each selected client $\mathcal{C}_i$ refines ω_t locally, obtaining updated parameters ω_t^i , and returns these to the server $\mathcal{S}$, which aggregates them into the new global model ω_{t+1} . This new global model is then broadcast for the next round of training. The process continues until convergence.

C. Fairness Evaluation Metrics

In this subsection, we present two independence metrics for the binary label, where $y \in \{0, 1\}$ denotes the ground truth label, $\hat{y} \in \{0, 1\}$ denotes the prediction label, and $s \in \{0, 1\}$ denotes the sensitive feature.

Definition 1 (Statistical Parity (i.e., Demographic Parity, Independence) [41]): Statistical parity requires the predictions to be independent of the sensitive features s . It can be formally written as:

$$\mathbb{P}(\hat{y}|s=0) = \mathbb{P}(\hat{y}|s=1). \quad (4)$$

When both the predicted labels and sensitive features are binary, to quantify the extent of statistical parity, the Δ_{SP} is defined as follows:

$$\Delta_{SP} = |\mathbb{P}(\hat{y}=1|s=0) - \mathbb{P}(\hat{y}=1|s=1)|. \quad (5)$$

The Δ_{SP} measures the acceptance rate difference between the two sensitive subgroups.

Definition 2 (Equal Opportunity [42]): Equal opportunity necessitates that the likelihood of an instance belonging to a positive class leading to a positive outcome should be equitable for all members within subgroups. For individuals with positive ground truth labels, it is necessary for positive predictions to be devoid of any dependence on sensitive features. This principle can be precisely expressed as follows:

$$\mathbb{P}(\hat{y}=1|y=1, s=0) = \mathbb{P}(\hat{y}=1|y=1, s=1). \quad (6)$$

Fairness-aware GNNs prevent the allocation of unfavourable predictions to individuals who are eligible for advantageous ones solely based on their sensitive subgroup affiliation. In particular, Δ_{EO} quantifies the extent of deviation in predictions from the ideal scenario where equality of opportunity is satisfied. In line with these considerations, we employ the following metrics to quantitatively assess equal opportunity:

$$\Delta_{EO} = |\mathbb{P}(\hat{y}=1|y=1, s=0) - \mathbb{P}(\hat{y}=1|y=1, s=1)|. \quad (7)$$

Both the probabilities are evaluated on the test set.

D. Global Fairness and Local Fairness

We consider both local fairness and global fairness in our evaluations. Let F^{Global} denote a global fairness evaluation, F_i^{Local} denote a fairness evaluation of client C_i and F^{Local} denote an overall local fairness evaluation. Specially, F^{Local} is influenced by the weight ω_i .

Definition 3 (Global Fairness Evaluation Metrics): Global fairness evaluation measures the fairness at the server-level, which is consistent with the fairness evaluation metric for graphs. Thus, F^{Global} can be $\Delta_{SP}^{\text{Global}}$ or $\Delta_{EO}^{\text{Global}}$.

Definition 4 (Local Fairness Evaluation Metrics): Local fairness evaluation measures the fairness in different clients. For each client C_i , ω^i denotes the final weight, and F_i^{Local} denotes the fairness evaluation. This is quantified using a weighted average, formally written as:

$$F^{\text{Local}} = \frac{\sum_{i=1}^K \omega^i F_i^{\text{Local}}}{\sum_{i=1}^K \omega^i}. \quad (8)$$

Specifically, in FedAvg framework, $\omega^i = \omega^j$ for all $i, j \in \{1, 2, \dots, K\}$. Thus, the local statistical parity is shown as:

$$F^{\text{Local}} = \frac{\sum_{i=1}^K F_i^{\text{Local}}}{K}. \quad (9)$$

Here, the F^{Local} can also be $\Delta_{SP}^{\text{Local}}$ or $\Delta_{EO}^{\text{Local}}$.

Lemma 1 ([8]): F^{Global} cannot be written as a linear combination of F_i^{Local} , $i \in 1, \dots, K$.

For FedGNNs, both client fairness and server fairness are equally important. Since these two metrics cannot be linearly transformed into each other, it is necessary to use both local fairness evaluations and global fairness evaluations to measure client fairness and server fairness, respectively.

E. Relevant Lemma

To understand the impact of graph convolutions on sensitive features, Luo et al. [26] analyze the transformation of sensitive feature channel h_s through multiple layers of graph convolution.

Lemma 2.1 ([26]): Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix with real-valued entries, and let $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$ denote its eigenvalues, with $\mathbf{p}_i$ ($i \in \{1, 2, \dots, n\}$) being corresponding eigenvectors. When the largest eigenvalue has a single corresponding eigenvector $\mathbf{p}_1$, the correlation between the sensitive features and the spectrum is primarily governed by $\mathbf{p}_1$. In this scenario, selecting $\mathbf{p}_1$ for spectral convolution ensures that the transformation of sensitive features is aligned with the principal geometric structure of the graph, thereby enhancing fairness while maintaining utility. Formally, this is represented by the following equation establishes:

$$\lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}^l h_s, h_s \rangle) = \cos(\langle h_s, \mathbf{p}_1 \rangle). \quad (10)$$

Given the eigendecomposition of $\mathbf{A}$ as $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^\top$ with $\mathbf{P} = (\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n)$, $\|\mathbf{p}_i\| = 1$ ($i \in \{1, 2, \dots, n\}$) and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. We assume $\beta_i = h_s^\top \mathbf{p}_i$, which is the weight of h_s on $\mathbf{p}_i$ when representing h_s with the set of orthonormal bases $\mathbf{p}_i$ ($i \in \{1, 2, \dots, n\}$). Thus,

$$\begin{aligned} & \lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}^l h_s, h_s \rangle) \\ &= \lim_{l \rightarrow \infty} \frac{(\mathbf{A}^l h_s)^\top h_s}{\|\mathbf{A}^l h_s\| \|h_s\|} \\ &= \lim_{l \rightarrow \infty} \frac{(\mathbf{A}^l h_s)^\top h_s}{\sqrt{(\mathbf{A}^l h_s)^\top \mathbf{A}^l h_s} \sqrt{h_s^\top h_s}} \\ &= \lim_{l \rightarrow \infty} \frac{(\mathbf{P}\mathbf{\Lambda}^l \mathbf{P}^\top h_s)^\top h_s}{\sqrt{(\mathbf{P}\mathbf{\Lambda}^l \mathbf{P}^\top h_s)^\top (\mathbf{P}\mathbf{\Lambda}^l \mathbf{P}^\top h_s)} \sqrt{h_s^\top h_s}} \\ &= \lim_{l \rightarrow \infty} \frac{(\mathbf{P}^\top h_s)^\top \mathbf{\Lambda}^l (\mathbf{P}^\top h_s)}{\sqrt{(\mathbf{P}^\top h_s)^\top \mathbf{\Lambda}^{2l} (\mathbf{P}^\top h_s)} \sqrt{h_s^\top h_s}} \\ &= \lim_{l \rightarrow \infty} \frac{\sum_{i=1}^n \beta_i^2 \lambda_i^{2l}}{\sqrt{\sum_{i=1}^n \beta_i^2 \lambda_i^{2l}} \sqrt{\sum_{i=1}^n \beta_i^2}} \\ &= \lim_{l \rightarrow \infty} \frac{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^{2l}}{\sqrt{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^{2l}} \sqrt{\sum_{i=1}^n \beta_i^2}}. \end{aligned} \quad (11)$$

Lemma 2.2 ([26]): In cases where the largest eigenvalue corresponds to multiple eigenvectors $\{\mathbf{p}_1, \dots, \mathbf{p}_j\}$, the correlation between sensitive features and the spectrum is influenced by all these eigenvectors. Here, the selection mechanism must consider all j eigenvectors to achieve a balanced trade-off between

fairness and utility. By incorporating these eigenvectors into the graph convolution process, graph convolution can mitigate the influence of non-principal eigenvectors, thereby preserving the independence of sensitive features and enhancing the overall fairness of the model. Formally, this is represented by the following equation:

$$\lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}^l h_s, h_s \rangle) \geq \frac{1}{\sqrt{j}} \sum_{i=1}^j \cos(\langle h_s, \mathbf{p}_i \rangle). \quad (12)$$

Lemma 2.3 ([26]): The correlation between non-principal eigenvectors $\mathbf{p}_i$ and $\cos(\langle \mathbf{S}^l h_s, h_s \rangle)$ is proportional to $(\frac{\lambda_i}{\lambda_1})^l$. This implies that the influence of non-principal eigenvectors on sensitive features decays exponentially over multiple layers of convolution. Formally, this is expressed as:

$$\begin{aligned} & \cos(\langle \mathbf{S}^l h_s, h_s \rangle) \\ &= \frac{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^l}{\sqrt{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^{2l}} \sqrt{\sum_{i=1}^n \beta_i^2}}. \end{aligned} \quad (13)$$

Theoretical analyses show that the correlation between sensitive features and the graph spectrum can be effectively managed through eigenvector selection. By leveraging the largest eigenvectors of the adjacency matrix, graph convolution enhances fairness without compromising utility, offering a robust framework for fair and efficient graph learning.

IV. THEORETICAL DISCOVERY

In this section, we present the theoretical findings underpinning F3GL. A critical aspect of harmonizing fairness and utility is understanding the correlation between sensitive features and the graph spectrum. Our mathematical analysis reveals that this correlation can be calculated using the largest eigenvector of the adjacency matrix. Furthermore, exploring the association of the client-level spectrum with two types of fairness is essential for implementing federated graph learning to achieve dual-perspective fairness. We theoretically analyze the relationship between local fairness and the client-level spectrum as well as the relationship between global fairness and the client-level spectrum.

A. The Decay of the Similarity

To address the gap in the rigorous mathematical proof regarding the exponential decrease in correlation, we present a detailed proof of the decay of the similarity between original sensitive features and those after l layers convolution.

Theorem 1: Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric matrix with real-valued entries. The eigenvalues of $\mathbf{A}$ are $|\lambda_1| > |\lambda_2| > |\lambda_3| \geq \dots \geq |\lambda_n|$. Then $\cos(\langle \mathbf{A}^l h_s, h_s \rangle)$ decays exponentially, and the rate of convergence depends only on the first two eigenvalues $|\lambda_1|, |\lambda_2|$.

Proof: Let $\mathbf{T}^l = \cos(\langle \mathbf{A}^l h_s, h_s \rangle)$, thus:

$$\mathbf{T}^l = \frac{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^l}{\sqrt{\beta_1^2 + \sum_{i=2}^n \beta_i^2 (\frac{\lambda_i}{\lambda_1})^{2l}} \cdot \sqrt{\sum_{i=1}^n \beta_i^2}}.$$

Let $\frac{\beta_i}{\beta_1} = u_i, \frac{\lambda_i}{\lambda_1} = v_i < 1$, thus:

$$\mathbf{T}^l = \frac{1 + \sum_{i=2}^n u_i^2 v_i^{2l}}{\sqrt{1 + \sum_{i=2}^n u_i^2 v_i^{2l}} \sqrt{1 + \sum_{i=2}^n u_i^2}}.$$

Let $\mathbf{T}' = \lim_{l \rightarrow \infty} \mathbf{T}^l$, thus:

$$\mathbf{T}' = \frac{\beta_1}{\sqrt{\sum_{i=1}^n \beta_i^2}} = \frac{1}{\sqrt{1 + \sum_{i=2}^n u_i^2}}.$$

Using the Taylor expansion of the multivariate function $f(\mathbf{x})$ (second order):

$$\begin{aligned} f(\mathbf{x}) &= f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top \\ &\quad \nabla^2 f(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0) + o(\|\mathbf{x} - \mathbf{x}_0\|)^2 \end{aligned}$$

Let $x_i = u_i v_i^l, f(x_2, \dots, x_n) = \sqrt{1 + \sum_{i=2}^n x_i^2}$. Thus,

$$\frac{\partial f}{\partial x_i} = \frac{1}{2} \cdot \frac{1 \cdot 2x_i}{\sqrt{1 + \sum_{i=2}^n x_i^2}} = \frac{x_i}{\sqrt{1 + \sum_{i=2}^n x_i^2}}.$$

When $x_2 = \dots = x_n = 0$,

$$\frac{\partial f}{\partial x_i} = 0$$

$$\begin{aligned} \frac{\partial f}{(\partial x_i)^2} &= \frac{\sqrt{1 + \sum_{i=2}^n x_i^2} \cdot 1 - x_i \cdot (-\frac{1}{2}) \cdot \frac{2x_i}{\sqrt{(1 + \sum_{i=2}^n x_i^2)^3}}}{1 + \sum_{i=2}^n x_i^2} \\ &= \frac{1}{\sqrt{1 + \sum_{i=2}^n x_i^2}} + \frac{x_i^2}{\sqrt{(1 + \sum_{i=2}^n x_i^2)^5}}. \end{aligned}$$

When $x_2 = \dots = x_n = 0$,

$$\frac{\partial f}{(\partial x_i)^2} = 1.$$

$$\frac{\partial f}{\partial x_i \partial x_j} = \frac{x_i x_j}{\sqrt{(1 + \sum_{i=2}^n x_i^2)^5}}.$$

When $x_2 = \dots = x_n = 0$,

$$\frac{\partial f}{\partial x_i \partial x_j} = 0.$$

Thus, the Taylor expansion of $f(x_2, \dots, x_n)$ at $(0, \dots, 0)$:

$$\begin{aligned} f(x_2, \dots, x_n) &= f(0, \dots, 0) + \sum_{i=2}^n (x_i - 0) f'_{x_i}(0, \dots, 0) \\ &\quad + \frac{1}{2} \sum_{i,j=2}^n (x_i - 0)(x_j - 0) f''_{x_i x_j}(0, \dots, 0) \\ &\quad + o^2 \\ &= 1 + 0 + \frac{1}{2} \sum_{i=2}^n x_i^2 + o^2 \\ &= 1 + \frac{1}{2} \sum_{i=2}^n x_i^2 + o^2. \end{aligned}$$

Thus:

$$\sqrt{1 + \sum_{i=2}^n (u_i v_i^l)^2} = 1 + \frac{1}{2} \sum_{i=2}^n (u_i v_i^l)^2 + o^2,$$

$$\sqrt{1 + \sum_{i=2}^n (u_i v_i^{l+1})^2} = 1 + \frac{1}{2} \sum_{i=2}^n (u_i v_i^{l+1})^2 + o^2.$$

Because $\lim_{l \rightarrow \infty} (1 + \frac{1}{2} \sum_{i=2}^n (u_i v_i^l)^2) = 1$, thus

$$\begin{aligned} \lim_{l \rightarrow \infty} \frac{\mathbf{T}^{l+1} - \mathbf{T}'}{\mathbf{T}^l - \mathbf{T}'} &= \lim_{l \rightarrow \infty} \frac{\sum_{i=2}^n u_i^2 v_i^{l+1} - \sum_{i=2}^n \frac{1}{2} u_i^2 v_i^{2l+2}}{\sum_{i=2}^n u_i^2 v_i^l - \sum_{i=2}^n \frac{1}{2} u_i^2 v_i^{2l}} \\ &= \lim_{l \rightarrow \infty} \frac{\sum_{i=2}^n u_i^2 (v_i^{l+1} - \frac{1}{2} v_i^{2l+2})}{\sum_{i=2}^n u_i^2 (v_i^l - \frac{1}{2} v_i^{2l})} \\ &= \lim_{l \rightarrow \infty} \frac{\sum_{i=2}^n u_i^2 v_i^{l+1}}{\sum_{i=2}^n u_i^2 v_i^l} \\ &= \lim_{l \rightarrow \infty} \frac{u_2^2 v_2^{l+1} + \sum_{i=3}^n u_i v_i^{l+1}}{u_2^2 v_2^l + \sum_{i=3}^n u_i v_i^l}. \end{aligned}$$

Let $\frac{v_i}{v_2} = \omega_i < 1$, thus:

$$\begin{aligned} \lim_{l \rightarrow \infty} \frac{\mathbf{T}^{l+1} - \mathbf{T}'}{\mathbf{T}^l - \mathbf{T}'} &= \lim_{l \rightarrow \infty} \frac{u_2^2 v_2 \cdot v_2^l + \sum_{i=3}^n u_i \omega_i^l v_i \cdot v_2^l}{u_2^2 \cdot v_2^l + \sum_{i=3}^n u_i \omega_i^l \cdot v_2^l} \\ &= \lim_{l \rightarrow \infty} \frac{u_2^2 v_2 + \sum_{i=3}^n u_i \omega_i^l v_i}{u_2^2 + \sum_{i=3}^n u_i \omega_i^l} \\ &= v_2 \\ &= \frac{\lambda_2}{\lambda_1}. \end{aligned}$$

Therefore, $\cos(\angle \mathbf{A}^l h_s, h_s)$ decays exponentially with the rate determined by $\frac{\lambda_2}{\lambda_1}$. ■

B. Global Fairness and Local Fairness

We discover that both global and local fairness are related to the maximum eigenvalue of the client adjacency matrix. To explore this, we make the following general assumptions: $V = V_1 \cup V_2 \dots \cup V_K$. And $\forall i, j, V_i \cap V_j = \emptyset$.

Lemma 3.1: If $i \neq j, \mathbf{A}'_i \mathbf{A}'_j = 0$.

Proof:

$$\forall i, j, \mathbf{A}'_i \mathbf{A}'_j[p, q] = \sum_{k=1}^n A'_i[p, k] A'_j[k, q].$$

Proof by contradiction. We propose the hypothesis:

$$\exists k, \mathbf{A}'_i[p, k] \mathbf{A}'_j[k, q] > 0.$$

Thus,

$$\mathbf{A}'_i[p, k] = \mathbf{A}'_j[k, q] = 1.$$

This implies $v_k \in \mathcal{V}_i$ and $v_k \in \mathcal{V}_j$, which contradicts the conditions: $\mathcal{V}_i \cap \mathcal{V}_j = \emptyset$. Therefore the hypothesis is not valid.

Hence,

$$\forall k, \mathbf{A}'_i[p, k] \mathbf{A}'_j[k, q] = 0.$$

Thus,

$$\forall p, q, \mathbf{A}'_i \mathbf{A}'_j[p, q] = 0.$$

Therefore,

$$\mathbf{A}'_i \mathbf{A}'_j = 0. \quad \blacksquare$$

Lemma 3.2: $\forall l, (\sum_{k=1}^K \mathbf{A}_k)^l = \sum_{k=1}^K \mathbf{A}_k^l$.

Proof: We use mathematical induction to proof this.

Base case ($K = 1$): $\sum_{k=1}^K \mathbf{A}_k = \mathbf{A}_1$. Thus,

$$\left(\sum_{k=1}^K \mathbf{A}_k \right)^l = (\mathbf{A}_1)^l = \mathbf{A}_1^l.$$

The conclusion is valid.

Inductive hypothesis ($K = i$): Assume that the conclusion holds when $k = i$, thus

$$\left(\sum_{k=1}^i \mathbf{A}_k \right)^l = \sum_{k=1}^i \mathbf{A}_k^l.$$

Inductive Step ($K = i + 1$):

$$\begin{aligned} &\left(\sum_{k=1}^{i+1} \mathbf{A}_k \right)^l \\ &= \left(\sum_{k=1}^i \mathbf{A}_k + \mathbf{A}_{i+1} \right)^l \\ &= \binom{l}{0} \mathbf{A}_{i+1}^l + \binom{l}{1} \mathbf{A}_{i+1}^{l-1} \sum_{k=1}^i \mathbf{A}_k + \dots + \binom{l}{l} \left(\sum_{k=1}^i \mathbf{A}_k \right)^l. \end{aligned}$$

Because, $\forall j \in \{1, 2 \dots i\}$:

$$j \neq i + 1.$$

Based on [Lemma 3.1](#),

$$\mathbf{A}_{i+1} \mathbf{A}_j = 0.$$

Then,

$$\begin{aligned} &\left(\sum_{k=1}^{i+1} \mathbf{A}_k \right)^l \\ &= \binom{l}{0} \mathbf{A}_{i+1}^l + \binom{l}{1} \left(\sum_{k=1}^i \mathbf{A}_k \right)^l \\ &= \mathbf{A}_{i+1}^l + \sum_{k=1}^i \mathbf{A}_k^l \\ &= \sum_{k=1}^{i+1} \mathbf{A}_k^l. \end{aligned}$$

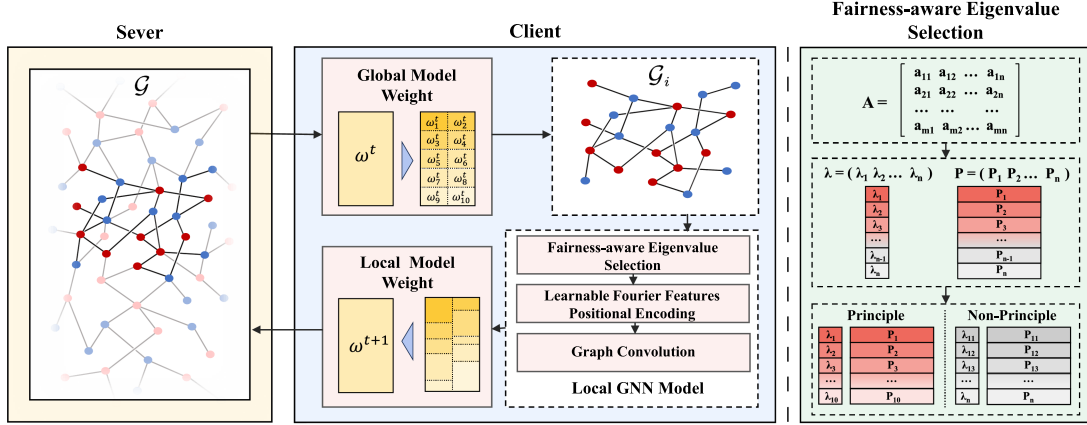


Fig. 2. The Framework of F3GL.

Combining the base case, the inductive hypothesis, and the inductive step, we show that this mathematical statement holds for all positive integer i .

Assume $\mathbf{A}' = \sum_{k=1}^K \mathbf{A}'_k$. The spectral graph convolution could be generalized as follows:

$$\mathbf{H}^{(l)} = (1 - \theta) \mathbf{A}' \mathbf{H}^{(l-1)} + \theta \mathbf{H}^{(0)}. \quad (14)$$

We denote the sensitive features in client $\mathcal{C}_i$ as $h_{s,i}$, the eigenvectors of $\mathbf{A}_k$ as $\{\mathbf{p}_{1,k}, \mathbf{p}_{2,k} \dots \mathbf{p}_{|V_k|,k}\}$ with the corresponding eigenvalues as $|\lambda_{1,k}| > |\lambda_{2,k}| \geq \dots \geq |\lambda_{|V_k|,k}|$, the eigenvectors of $\mathbf{A}'_k$ as $\{\mathbf{p}'_{1,k}, \mathbf{p}'_{2,k} \dots \mathbf{p}'_{n,k}\}$ with corresponding eigenvalues $|\lambda'_{1,k}| > |\lambda'_{2,k}| \geq \dots \geq |\lambda'_{n,k}|$.

Theorem 2 (Local Fairness): For each client $\mathcal{C}_i$, we have:

$$\lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}_k^l h_{s,i}, h_{s,i} \rangle) = \cos(\langle h_{s,i}, \mathbf{p}_{1,i} \rangle). \quad (15)$$

Thus, the local fairness is correlated with all the eigenvectors corresponding to the largest magnitude eigenvalue in each client:

$$\begin{aligned} & \lim_{l \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \cos(\langle \mathbf{A}_k^l h_{s,i}, h_{s,i} \rangle) \\ &= \frac{1}{K} \sum_{k=1}^K \cos(\langle h_{s,i}, \mathbf{p}_{1,i} \rangle). \end{aligned} \quad (16)$$

The proof is similar with Lemma 2.1.

Theorem 3 (Global Fairness): For the server, we have:

$$\lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}^l h_s, h_s \rangle) \geq \min_{1 \leq k \leq K} \{\cos(\langle h_s, \mathbf{p}'_{1,k} \rangle)\}. \quad (17)$$

Proof:

$$\begin{aligned} & \lim_{l \rightarrow \infty} \cos(\langle \mathbf{A}^l h_s, h_s \rangle) \\ &= \lim_{l \rightarrow \infty} \cos\left(\left\langle \sum_{k=1}^K \mathbf{A}_k^l h_s, h_s \right\rangle\right) \\ &= \lim_{l \rightarrow \infty} \frac{\sum_{k=1}^K \mathbf{A}_k^l h_s h_s}{\|\sum_{k=1}^K \mathbf{A}_k^l h_s\| \|h_s\|} \end{aligned}$$

$$\begin{aligned} &= \lim_{l \rightarrow \infty} \sum_{k=1}^K \frac{\|\mathbf{A}_k^l h_s\|}{\|\sum_{k=1}^K \mathbf{A}_k^l h_s\|} \frac{\mathbf{A}_k^l h_s}{\|\mathbf{A}_k^l h_s\|} \\ &= \lim_{l \rightarrow \infty} \sum_{k=1}^K \frac{\|\mathbf{A}_k^l h_s\|}{\|\sum_{k=1}^K \mathbf{A}_k^l h_s\|} \cos(\langle \mathbf{A}_k^l h_s, h_s \rangle) \\ &\geq \lim_{l \rightarrow \infty} \sum_{k=1}^K \frac{\|\mathbf{A}_k^l h_s\|}{\sum_{k=1}^K \|\mathbf{A}_k^l h_s\|} \cos(\langle \mathbf{A}_k^l h_s, h_s \rangle). \end{aligned}$$

Because,

$$\sum_{k=1}^K \frac{\|\mathbf{A}_k^l h_s\|}{\sum_{k=1}^K \|\mathbf{A}_k^l h_s\|} = 1.$$

Thus,

$$\begin{aligned} & \lim_{l \rightarrow \infty} \sum_{k=1}^K \frac{\|\mathbf{A}_k^l h_s\|}{\sum_{k=1}^K \|\mathbf{A}_k^l h_s\|} \cos(\langle \mathbf{A}_k^l h_s, h_s \rangle) \\ &\geq \min_{1 \leq k \leq K} \{\lim_{l \rightarrow \infty} \{\cos(\langle \mathbf{A}_k^l h_s, h_s \rangle)\}\} \\ &= \min_{1 \leq k \leq K} \{\cos(\langle h_s, \mathbf{p}'_{1,k} \rangle)\}. \end{aligned}$$

This demonstrates that both local and global similarity between the original sensitive features and those after l layers of convolution in federated learning can be calculated or bounded by the eigenvectors corresponding to the largest magnitude eigenvalue. Therefore, fairness-aware eigenvalue selection, coupled with adjustments to the selected eigenvectors, ensures fairness of spectral convolution.

V. THE DESIGN OF F3GL

In this section, we introduce the technical details of F3GL. Firstly, F3GL follows the principles of FedAvg. In each round t , the sever $\mathcal{S}$ uniformly samples a set of clients. Secondly, for each client, we calculate the e largest magnitude eigenvalues and their corresponding eigenvectors of the adjacency matrix.

The largest magnitude eigenvalues effectively express the graph structure and maintain the utility of the algorithm [43]. Based on Theorems 1, 2, and 3, excluding non-principal eigenvectors also improves fairness. Thirdly, we adjust the distribution of eigenvectors using learnable Fourier features positional encoding [44], which further enhances the algorithm's utility. Lastly, we incorporate the optimized spectrum into federated graph convolution. By integrating these steps, F3GL ensures both fairness and utility in federated learning settings, addressing key challenges in FedGNNs. Fig. 2 illustrates the model architecture.

A. Federated Learning Framework

In the central server, the simple and efficient FedAvg algorithm is often used as the optimization method, weighting the updated local averaging parameters based on the different data sizes of the client-side data [40].

After initializing the weight ω^0 of the global model, given K clients, the process of federated averaging in one communication round includes the following steps: The server selects a subset of clients $\mathcal{S} = (\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K)$. Each client graph $\mathcal{C}_k$ has its private graph $\mathcal{G}_k$ of each client. The server distributes the current weight ω^t to $\mathcal{C}_k$. The selected clients train the model with their local data, updating the local model weight ω_k^t to ω_k^{t+1} using the following local training process:

$$\omega_k^{t+1} = \omega^t - \eta \mathcal{L}(\omega_k^t, \mathcal{G}_k), \quad (18)$$

where $\mathcal{L}$ is the loss function, and η indicates the learning rate. Clients upload the locally trained weight ω_k^{t+1} to the server. The server performs an averaging process of all received local model weights to output the updated global weight ω^{t+1} :

$$\omega^{t+1} = \sum_{k=1}^K \frac{n_k}{\sum_{k \in K} n_k} \omega_k^{t+1}. \quad (19)$$

where n_k is the data size of client k .

Under the aforementioned federated learning settings, we can train a global prediction model based on the local datasets.

B. Fairness-Aware Eigenvalue Selection

For client $\mathcal{C}_i$, as the dimension of feature space expands, the influence of eigenvectors on the sensitive feature channel increases. Selecting a subset of eigenvalues reduces this influence and enhances the independence of sensitive features. We confirm that, after l layers convolution, sensitive features correlate with the largest magnitude eigenvalue and its corresponding eigenvectors. Utilizing the Arnoldi Package algorithm [45], it becomes feasible to represent the eigenvalues of the adjacency matrix by calculating a subset of e eigenvalues and corresponding eigenvectors:

$$\begin{aligned} \lambda_{\text{FES}} &= (\lambda_1, \lambda_2, \dots, \lambda_e)^\top, \\ \mathbf{p}_{\text{FES}} &= (\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_e)^\top. \end{aligned} \quad (20)$$

We choose the Arnoldi Package due to its high computational efficiency and accuracy, as empirically demonstrated in [46]. Here, e is a model hyperparameter representing the number of principal eigenvectors retained in the graph spectrum. In our

empirical analysis in Section IV, we show that the effectiveness of the model (in terms of fairness and utility) is influenced by the value of e . At lower values of e (e.g., $e < 10$), the model maintains high effectiveness with only slight fluctuations. However, when too many non-principal eigenvectors are included in the spectrum (e.g., $e > 100$), the model's effectiveness decreases significantly.

This strategic selection of eigenvalues and eigenvectors, facilitated by the Arnoldi Package, ensures that the sensitive features maintain a fair representation in the FedGNNs, ultimately improving both the fairness and utility of the model.

C. Learnable Fourier Features Positional Encoding

We use the learnable Fourier feature positional encoding function [44] that maps a 1-dimensional position $\lambda_i \in \mathbb{R}^1, i \in \{1, 2, \dots, e\}$ into a D_p -dimensional feature vector. This D_p -dimensional vector will then be used in downstream computation for attention mechanisms. The proposed encoding function is composed of the following components.

To extract useful features from the input position λ_i , the learnable Fourier feature positional encoding considers the following feature extraction layer motivated by the idea of Fourier features. Given a 1-dimensional position $\lambda_i \in \mathbb{R}^1, i \in \{1, 2, \dots, e\}$, we acquire a D_f -dimensional Fourier feature vector representation for the position, $\lambda_{\text{PE}}^i \in \mathbb{R}^{D_f}$, as follows:

$$\lambda_{\text{PE},1}^i = \frac{1}{\sqrt{D_f}} [\cos \lambda_i W_1^T \parallel \sin \lambda_i W_1^T], i \in \{1, 2, \dots, e\}, \quad (21)$$

where $\parallel$ denotes the concatenation of two vectors. This can also be viewed as the generalization of sinusoidal position encoding to the multi-dimensional case. Here, we set $W_1 \in \mathbb{R}^{\frac{D_f}{2} \times D_p}$, which defines both the orientation and wavelength of Fourier features, as trainable parameters.

To feed the representation to the downstream computation, we enhance the representation's capacity by modulating the features with a multi-layer perceptron (MLP):

$$\lambda_{\text{PE},2}^i = \text{GeLU}(\lambda_{\text{PE},1}^i W_2 + B_1) W_3 + B_2, \quad (22)$$

where $W_2 \in \mathbb{R}^{D_f \times D_h}$ and $W_3 \in \mathbb{R}^{D_h \times D_p}$ are initialized learnable weights, D_h is the hidden layer dimension, B_1 and B_2 are biases.

Finally, we reshape $\lambda_{\text{PE},2}^i$ into the shape of $[1, D_p]$, which is the λ_{PE}^i . We denote $\lambda_{\text{PE}} = (\lambda_{\text{PE}}^1, \dots, \lambda_{\text{PE}}^e)$. This component ensures that the positional encoding is both flexible and learnable, allowing the model to dynamically adjust the positional representations according to the task. More importantly, by applying learnable positional encoding—specifically Fourier-based encoding—we aim to regularize the distribution of the selected eigenvectors, which helps mitigate the directional inconsistency introduced by eigenvalue selection [26]. Such regularization not only enhances the utility of the graph convolution operation but also contributes to improved fairness.

TABLE III
STATISTICS OF POKEC-Z AND POKEC-N DATASETS

Dataset	# Nodes	# Edges	Sensitive Feature	Label
Credit	30,000	137,377	Age	Credit
Pokec-z	67,797	882,765	Region/Gender	Field
Pokec-n	66,569	729,129	Region/Gender	Field
AMiner-L	129,726	591,039	Affiliation	Field

D. Graph Convolution

Finally, the proposed method uses MLP with layer normalization (LN) as follows:

$$\lambda_{\text{MLP}} = \text{MLP}(\text{LN}(\lambda_{\text{PE}})) + \lambda_{\text{PE}}. \quad (23)$$

We optimize the spectrum based on learned basis λ_{MLP} and transform the $\mathbf{H}^{(l-1)}$ into:

$$\mathbf{H}'^{(l-1)} = \mathbf{P}_{\text{FES}} \cdot (\lambda_{\text{MLP}} \odot \mathbf{P}_{\text{FES}}^{\top} \mathbf{H}^{(l-1)}). \quad (24)$$

where $\cdot$ denotes the element-wise multiplication.

The graph convolution can then be written as follows:

$$\mathbf{H}^{(l)} = \sigma((\mathbf{H}^{(l-1)} \parallel \mathbf{H}'^{(l-1)}) \mathbf{W}_4^{(l-1)}), \quad (25)$$

where $\mathbf{W}_4^{(l-1)}$ is the transformation matrix, $\parallel$ denotes concatenation, and σ is the activation function. By stacking multiple graph convolutional layers, F3GL can learn node representations in each client. This approach ensures that the spectral information and positional encoding are effectively leveraged, enhancing both the fairness and utility of FedGNNs.

VI. EXPERIMENTS

A. Datasets

We demonstrate the effectiveness of our method on the downstream task of node classification using four real-world datasets: **Credit**, **Pokec-z**, **Pokec-n** and **Aminer-L**.

Credit is based on similarities in the users' spending and payment patterns. The sensitive feature is Age, while the label indicates whether an individual defaults on credit payments in the following month. Pokec-z and Pokec-n are derived from **Pokec**, a popular social network in Slovakia [47]. Pokec-z and Pokec-n are constructed by sampling users from two provinces, Zilinsky and Nitriansky, respectively. In both datasets, each user corresponds to a node, and edges represent friendship relations between users. The feature vector of each node is generated based on the user's profile, including attributes such as living region, gender, spoken language, and age. In these datasets, the users' living region and gender are considered sensitive features. The classification task involves predicting the working field of users. We denote the datasets where the living region is the sensitive feature as Pokec-z-R and Pokec-n-R, respectively. Similarly, we denote the datasets where gender is the sensitive feature as Pokec-z-G and Pokec-n-G. AMiner-L is a co-author network constructed from the AMiner network. The continent of each researcher's affiliation is considered the sensitive feature. The target of downstream is to predict the primary research

field of each researcher. The statistics of these datasets are summarized in Table III.

For datasets with more than two classes of ground truth labels for the node classification task, we simplify the problem by keeping the classes 0 and 1 unchanged, while setting the classes of labels greater than 1 to 1. This simplification helps balance the classes and make the task more manageable. We partition the data by randomly selecting 25% of nodes as the validation set and another 25% as the test set, ensuring that each category of nodes remains balanced in these sets. For the training set, we randomly select either 50% of nodes or 500 nodes in each class of ground truth labels, depending on which is smaller. This partitioning strategy is consistent with the approach used in several prior studies [48], [49], [50]. By maintaining a balanced representation of classes in the training, validation, and test sets, we ensure that the model is evaluated fairly and consistently across different partitions. This stratified partitioning ensures a fair and consistent evaluation of the proposed F3GL method against baselines.

B. Baselines

We compare our proposed method with eleven representative baselines from four categories: spectral convolutional network, fairness-aware GNNs, FedGNNs, and fair federated learning methods. To adapt non-federated graph learning methods for federated settings, we utilize three unified federated learning framework, FedAvg, FedNova [51] and FedProx [52].

GCN [27], **SAGE** [53], and **APNP** [37] are spectral convolutional networks that aim to optimize model utility. GCN leverages the spectral domain of graphs, using graph eigenvalues and eigenvectors to define convolution operations for node-level information aggregation and learning. GraphSAGE utilizes the graph Laplacian eigenvalues and eigenvectors to generate node embeddings by aggregating features from neighbouring nodes across multiple hops. APNP is a spectral graph neural network model that incorporates personalized PageRank information propagation to improve semi-supervised node classification on graphs. In addition, **GDC** [54] serves as a pre-processing technique for GCNs to highlight informative connections, whereas **GAT** [55] offers greater flexibility in controlling the aggregation of node information than traditional GCNs.

We adopt four state-of-the-art fairness-aware GNNs with the federated learning framework: **NIFTY** [48] seeks to optimize the alignment between predictions derived from perturbed sensitive features and those generated using unperturbed features. **FairVGNN** [20] designs two feature masking strategies according to feature correlations to highlight the importance of considering feature propagation and correlation variation in alleviating discrimination. **FairGNN** [23] utilizes adversarial training to eliminate sensitive feature information from node embeddings. **FairSIN** [24] unveils an innovative neutralization-based paradigm, where additional fairness-facilitating features are incorporated into node features or representations before message passing. **FED-PUB** [14] and **FedGCN** [13] are two state-of-the-art FedGNNs. FED-PUB focuses on the joint improvement

TABLE IV

COMPARISON ON UTILITY (ACC AND AUC) AND FAIRNESS (Δ_{SP} AND Δ_{EO}) IN PERCENTAGE (%) WITH POKEC-Z AND POKEC-N. $\uparrow$ DENOTES THE LARGER, THE BETTER; $\downarrow$ DENOTES THE OPPOSITE. THE BEST RESULTS ARE RED AND BOLD-FACED. THE RUNNER-UPS ARE BLUE AND UNDERLINED.

Methods	Pokec-z-R, Local				Pokec-z-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+FL}	70.85 $\pm$ 0.45	73.52 $\pm$ 0.85	23.35 $\pm$ 1.87	17.31 $\pm$ 1.32	68.49 $\pm$ 1.84	73.37 $\pm$ 0.98	14.91 $\pm$ 1.43	9.52 $\pm$ 2.31
SAGE _{+FL}	69.30 $\pm$ 1.64	72.98 $\pm$ 1.58	24.20 $\pm$ 2.06	21.50 $\pm$ 1.52	67.13 $\pm$ 1.02	73.11 $\pm$ 1.28	14.49 $\pm$ 0.42	10.42 $\pm$ 1.45
APPNP _{+FL}	69.44 $\pm$ 0.48	74.11 $\pm$ 1.84	25.02 $\pm$ 3.04	22.41 $\pm$ 1.63	67.24 $\pm$ 2.01	74.08 $\pm$ 2.19	17.81 $\pm$ 2.01	10.73 $\pm$ 0.84
GDC _{+FL}	68.85 $\pm$ 1.46	68.45 $\pm$ 1.72	22.35 $\pm$ 1.98	21.20 $\pm$ 1.54	66.70 $\pm$ 1.96	72.35 $\pm$ 1.85	10.70 $\pm$ 1.66	8.73 $\pm$ 1.45
GAT _{+FL}	69.26 $\pm$ 1.78	71.47 $\pm$ 0.87	21.88 $\pm$ 1.28	20.54 $\pm$ 1.33	66.98 $\pm$ 2.95	72.91 $\pm$ 0.54	11.84 $\pm$ 1.86	8.44 $\pm$ 1.00
NIFTY _{+FL}	67.66 $\pm$ 1.72	70.02 $\pm$ 2.53	19.42 $\pm$ 1.68	17.66 $\pm$ 1.98	65.47 $\pm$ 2.13	71.37 $\pm$ 1.40	8.06 $\pm$ 1.54	4.11 $\pm$ 0.95
FairVGNN _{+FL}	64.35 $\pm$ 1.17	68.55 $\pm$ 0.29	28.18 $\pm$ 0.01	28.47 $\pm$ 0.54	63.87 $\pm$ 0.10	69.85 $\pm$ 0.20	22.67 $\pm$ 0.65	16.86 $\pm$ 0.80
FairGNN _{+FL}	63.66 $\pm$ 2.10	71.41 $\pm$ 1.43	19.50 $\pm$ 1.87	<u>14.18 $\pm$ 0.82</u>	61.07 $\pm$ 0.89	72.41 $\pm$ 0.93	21.91 $\pm$ 2.06	14.28 $\pm$ 2.09
FairSIN _{+FL}	68.46 $\pm$ 1.42	70.13 $\pm$ 0.97	31.77 $\pm$ 2.98	27.19 $\pm$ 2.46	66.60 $\pm$ 0.89	71.67 $\pm$ 1.98	25.43 $\pm$ 1.46	20.14 $\pm$ 0.98
FED-PUB	68.17 $\pm$ 2.73	72.00 $\pm$ 1.79	22.71 $\pm$ 2.40	20.77 $\pm$ 4.00	65.94 $\pm$ 2.75	73.30 $\pm$ 1.05	11.47 $\pm$ 4.41	7.77 $\pm$ 3.83
FedGCN	69.70 $\pm$ 1.37	73.06 $\pm$ 1.97	24.66 $\pm$ 1.18	21.97 $\pm$ 2.79	67.34 $\pm$ 0.54	72.88 $\pm$ 0.09	15.24 $\pm$ 2.14	9.72 $\pm$ 0.30
FairFed	59.44 $\pm$ 3.60	63.70 $\pm$ 2.13	20.12 $\pm$ 1.32	18.69 $\pm$ 1.69	65.48 $\pm$ 2.63	69.72 $\pm$ 2.70	2.92 $\pm$ 1.28	<u>3.08 $\pm$ 1.21</u>
F ² GNN	68.43 $\pm$ 0.99	72.74 $\pm$ 2.31	<u>16.50 $\pm$ 0.78</u>	16.31 $\pm$ 0.28	67.71 $\pm$ 0.19	73.40 $\pm$ 0.39	<u>2.66 $\pm$ 1.02</u>	4.49 $\pm$ 0.25
F3GL	<u>69.85 $\pm$ 0.42</u>	<u>73.62 $\pm$ 0.67</u>	14.87 $\pm$ 2.27	13.37 $\pm$ 1.16	<u>67.93 $\pm$ 0.46</u>	<u>73.68 $\pm$ 0.76</u>	0.32 $\pm$ 0.02	0.47 $\pm$ 0.33

Methods	Pokec-n-R, Local				Pokec-n-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+FL}	69.67 $\pm$ 1.54	68.81 $\pm$ 0.44	20.97 $\pm$ 1.78	20.82 $\pm$ 1.12	68.14 $\pm$ 0.48	<u>71.93 $\pm$ 0.87</u>	3.31 $\pm$ 0.84	1.69 $\pm$ 0.48
SAGE _{+FL}	68.14 $\pm$ 1.46	68.06 $\pm$ 0.85	19.04 $\pm$ 0.69	20.90 $\pm$ 0.35	66.82 $\pm$ 0.12	70.91 $\pm$ 0.38	3.14 $\pm$ 1.24	<u>1.47 $\pm$ 0.54</u>
APPNP _{+FL}	<u>68.57 $\pm$ 1.84</u>	68.52 $\pm$ 0.49	<u>12.93 $\pm$ 1.34</u>	20.66 $\pm$ 1.26	67.23 $\pm$ 1.02	71.65 $\pm$ 0.34	7.08 $\pm$ 1.18	4.93 $\pm$ 0.23
GDC _{+FL}	67.91 $\pm$ 1.83	68.77 $\pm$ 1.65	17.72 $\pm$ 2.51	20.52 $\pm$ 2.16	67.51 $\pm$ 2.74	70.78 $\pm$ 1.20	2.40 $\pm$ 0.78	2.46 $\pm$ 0.72
GAT _{+FL}	67.86 $\pm$ 1.10	68.03 $\pm$ 1.17	18.54 $\pm$ 0.30	22.09 $\pm$ 2.93	66.93 $\pm$ 1.11	70.44 $\pm$ 2.13	3.06 $\pm$ 0.72	1.75 $\pm$ 0.84
NIFTY _{+FL}	65.88 $\pm$ 0.31	64.41 $\pm$ 0.70	19.09 $\pm$ 1.00	22.38 $\pm$ 0.90	65.59 $\pm$ 0.21	69.60 $\pm$ 0.20	7.43 $\pm$ 0.64	9.54 $\pm$ 0.20
FairVGNN _{+FL}	65.67 $\pm$ 1.27	65.90 $\pm$ 1.35	16.18 $\pm$ 1.00	18.48 $\pm$ 0.43	62.95 $\pm$ 0.26	68.98 $\pm$ 0.27	2.84 $\pm$ 0.07	4.02 $\pm$ 0.55
FairGNN _{+FL}	66.75 $\pm$ 2.17	67.44 $\pm$ 0.29	17.77 $\pm$ 0.01	20.66 $\pm$ 0.54	65.18 $\pm$ 0.10	70.34 $\pm$ 0.20	3.71 $\pm$ 0.65	2.36 $\pm$ 0.80
FairSIN _{+FL}	65.46 $\pm$ 1.00	67.48 $\pm$ 1.56	17.91 $\pm$ 2.08	18.74 $\pm$ 0.08	64.00 $\pm$ 0.17	70.40 $\pm$ 0.19	<u>1.17 $\pm$ 0.25</u>	1.57 $\pm$ 0.40
FED-PUB	68.02 $\pm$ 0.18	69.42 $\pm$ 0.34	18.55 $\pm$ 1.25	24.14 $\pm$ 1.13	66.80 $\pm$ 0.10	71.69 $\pm$ 0.19	9.21 $\pm$ 2.71	10.21 $\pm$ 1.08
FedGCN	68.52 $\pm$ 0.28	68.96 $\pm$ 1.01	18.20 $\pm$ 1.29	22.39 $\pm$ 1.75	67.27 $\pm$ 0.09	70.37 $\pm$ 0.68	2.97 $\pm$ 0.14	1.54 $\pm$ 0.62
FairFed	56.26 $\pm$ 2.42	58.41 $\pm$ 1.75	18.38 $\pm$ 2.03	18.79 $\pm$ 1.63	61.16 $\pm$ 2.46	61.97 $\pm$ 1.93	2.66 $\pm$ 1.20	3.69 $\pm$ 2.95
F ² GNN	67.75 $\pm$ 1.71	68.48 $\pm$ 0.81	16.38 $\pm$ 0.72	<u>16.23 $\pm$ 2.51</u>	67.00 $\pm$ 0.27	69.62 $\pm$ 1.43	1.58 $\pm$ 0.31	2.00 $\pm$ 1.11
F3GL	68.38 $\pm$ 1.08	<u>69.00 $\pm$ 2.05</u>	12.36 $\pm$ 1.39	14.41 $\pm$ 2.00	<u>67.59 $\pm$ 0.82</u>	71.99 $\pm$ 1.31	0.65 $\pm$ 0.42	1.03 $\pm$ 0.16

Methods	Pokec-z-G, Local				Pokec-z-G, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+FL}	69.70 $\pm$ 0.93	73.07 $\pm$ 0.73	8.81 $\pm$ 0.92	6.62 $\pm$ 0.92	67.34 $\pm$ 0.92	73.88 $\pm$ 0.73	6.68 $\pm$ 2.57	2.31 $\pm$ 0.61
SAGE _{+FL}	71.10 $\pm$ 0.18	74.16 $\pm$ 0.13	12.10 $\pm$ 1.40	9.24 $\pm$ 0.41	68.94 $\pm$ 0.23	75.14 $\pm$ 0.19	11.21 $\pm$ 1.56	7.02 $\pm$ 1.17
APPNP _{+FL}	66.95 $\pm$ 0.69	72.64 $\pm$ 1.13	8.50 $\pm$ 1.62	6.87 $\pm$ 0.60	66.91 $\pm$ 0.60	67.81 $\pm$ 1.03	6.17 $\pm$ 2.40	3.44 $\pm$ 1.90
GDC _{+FL}	69.64 $\pm$ 0.14	73.06 $\pm$ 0.09	5.09 $\pm$ 0.78	5.89 $\pm$ 0.16	67.21 $\pm$ 0.07	73.84 $\pm$ 0.05	3.96 $\pm$ 1.45	4.74 $\pm$ 1.64
GAT _{+FL}	69.27 $\pm$ 2.69	71.46 $\pm$ 0.88	5.39 $\pm$ 1.85	5.37 $\pm$ 0.21	66.99 $\pm$ 2.89	72.91 $\pm$ 0.54	4.98 $\pm$ 0.73	4.03 $\pm$ 0.72
NIFTY _{+FL}	65.54 $\pm$ 1.65	68.19 $\pm$ 1.42	4.93 $\pm$ 0.72	6.84 $\pm$ 1.23	63.25 $\pm$ 1.72	66.13 $\pm$ 1.13	2.66 $\pm$ 0.86	3.83 $\pm$ 1.59
FairVGNN _{+FL}	65.57 $\pm$ 0.95	68.96 $\pm$ 0.34	6.54 $\pm$ 1.26	7.61 $\pm$ 0.94	65.45 $\pm$ 0.60	64.58 $\pm$ 0.52	<u>2.40 $\pm$ 0.36</u>	5.46 $\pm$ 1.09
FairGNN _{+FL}	68.45 $\pm$ 2.43	68.85 $\pm$ 2.62	4.61 $\pm$ 1.64	5.16 $\pm$ 0.60	66.21 $\pm$ 2.21	66.21 $\pm$ 2.28	5.39 $\pm$ 2.48	4.62 $\pm$ 2.32
FairSIN _{+FL}	65.89 $\pm$ 0.96	69.16 $\pm$ 0.42	5.96 $\pm$ 1.49	5.22 $\pm$ 0.72	65.91 $\pm$ 1.04	65.12 $\pm$ 0.39	6.00 $\pm$ 2.49	5.62 $\pm$ 1.54
FED-PUB	68.15 $\pm$ 2.74	72.01 $\pm$ 1.80	7.31 $\pm$ 1.24	<u>4.91 $\pm$ 0.90</u>	65.91 $\pm$ 2.75	73.31 $\pm$ 1.06	4.91 $\pm$ 2.40	3.21 $\pm$ 0.98
FedGCN	69.70 $\pm$ 0.55	73.10 $\pm$ 0.04	5.93 $\pm$ 0.54	5.90 $\pm$ 0.28	67.89 $\pm$ 0.55	73.76 $\pm$ 0.49	3.61 $\pm$ 0.34	3.29 $\pm$ 0.02
FairFed	61.70 $\pm$ 1.24	63.79 $\pm$ 2.05	6.08 $\pm$ 1.30	6.97 $\pm$ 1.36	63.63 $\pm$ 1.76	67.31 $\pm$ 2.39	3.50 $\pm$ 0.80	4.05 $\pm$ 0.59
F ² GNN	66.54 $\pm$ 1.34	69.58 $\pm$ 1.26	<u>4.23 $\pm$ 1.04</u>	5.61 $\pm$ 0.06	65.13 $\pm$ 1.12	68.21 $\pm$ 1.82	3.39 $\pm$ 1.84	3.26 $\pm$ 1.23
F3GL	<u>70.09 $\pm$ 0.25</u>	<u>73.13 $\pm$ 0.19</u>	3.90 $\pm$ 0.72	3.72 $\pm$ 1.12	<u>68.20 $\pm$ 0.12</u>	<u>74.54 $\pm$ 0.21</u>	2.07 $\pm$ 0.64	<u>2.98 $\pm$ 1.03</u>

Methods	Pokec-n-G, Local				Pokec-n-G, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+FL}	71.75 $\pm$ 0.60	72.96 $\pm$ 0.79	5.85 $\pm$ 0.68	8.03 $\pm$ 1.31	67.34 $\pm$ 0.51	73.88 $\pm$ 0.69	7.32 $\pm$ 1.35	12.03 $\pm$ 1.68
SAGE _{+FL}	69.21 $\pm$ 0.58	72.13 $\pm$ 0.41	10.34 $\pm$ 1.64	7.35 $\pm$ 1.23	<u>67.37 $\pm$ 0.57</u>	74.73 $\pm$ 0.25	11.69 $\pm$ 1.57	<u>6.34 $\pm$ 2.00</u>
APPNP _{+FL}	71.47 $\pm$ 0.40	72.64 $\pm$ 0.73	<u>4.70 $\pm$ 0.42</u>	9.77 $\pm$ 1.66	66.91 $\pm$ 0.41	73.79 $\pm$ 0.52	8.83 $\pm$ 3.33	17.14 $\pm$ 3.97
GDC _{+FL}	70.91 $\pm$ 1.63	72.68 $\pm$ 1.35	5.85 $\pm$ 0.19	7.33 $\pm$ 0.31	66.55 $\pm$ 1.33	71.09 $\pm$ 2.34	8.41 $\pm$ 0.67	6.80 $\pm$ 0.22
GAT _{+FL}	70.42 $\pm$ 1.14	72.63 $\pm$ 1.35	5.07 $\pm$ 2.06	8.04 $\pm$ 1.52	66.78 $\pm$ 1.11	<u>74.24 $\pm$ 1.10</u>	7.21 $\pm$ 0.55	7.47 $\pm$ 0.56
NIFTY _{+FL}	65.89 $\pm$ 2.18	69.16 $\pm$ 3.26	5.96 $\pm$ 0.77	5.22 $\pm$ 1.65	63.25 $\pm$ 1.89	70.69 $\pm$ 2.12	11.30 $\pm$ 0.62	20.17 $\pm$ 1.60
FairVGNN _{+FL}	65.49 $\pm$ 1.09	67.84 $\pm$ 1.03	8.64 $\pm$ 1.15	12.10 $\pm$ 1.75	65.45 $\pm$ 1.05	70.11 $\pm$ 0.57	14.79 $\pm$ 0.99	17.06 $\pm$ 0.95
FairGNN _{+FL}	66.45 $\pm$ 0.07	68.85 $\pm$ 0.11	6.15 $\pm$ 0.73	8.05 $\pm$ 1.23	66.21 $\pm$ 0.01	70.34 $\pm$ 0.27	<u>6.61 $\pm$ 3.19</u>	12.29 $\pm$ 2.87
FairSIN _{+FL}	67.01 $\pm$ 1.39	68.13 $\pm$ 1.46	9.73 $\pm$ 2.23	13.72 $\pm$ 1.93	65.91 $\pm$ 1.50	70.42 $\pm$ 1.69	15.05 $\pm$ 3.05	19.07 $\pm$ 2.42
FED-PUB	68.59 $\pm$ 1.62	71.08 $\pm$ 2.08	7.36 $\pm$ 1.55	11.21 $\pm$ 2.39	67.18 $\pm$ 1.64	73.35 $\pm$ 2.05	7.64 $\pm$ 1.72	16.21 $\pm$ 2.32
FedGCN	<u>71.53 $\pm$ 1.32</u>	<u>72.96 $\pm$ 1.01</u>	5.93 $\pm$ 0.84	6.66 $\pm$ 1.65	67.27 $\pm$ 1.04	73.37 $\pm$ 1.56	6.69 $\pm$ 0.62	12.67 $\pm$ 0.64
FairFed	58.62 $\pm$ 4.24	62.36 $\pm$ 0.97	6.82 $\pm$ 1.30	10.97 $\pm$ 1.36	64.40 $\pm$ 1.64	70.37 $\pm$ 0.39	7.34 $\pm$ 2.01	13.30 $\pm$ 2.59
F ² GNN	67.45 $\pm$ 0.70	68.50 $\pm$ 1.25	5.51 $\pm$ 1.37	7.05 $\pm$ 0.21	66.67 $\pm$ 1.00	70.34 $\pm$ 0.27	6.93 $\pm$ 0.91	8.92 $\pm$ 0.78
F3GL	68.13 $\pm$ 0.17	69.44 $\pm$ 0.16	3.60 $\pm$ 0.74	<u>6.27 $\pm$ 1.43</u>	67.41 $\pm$ 0.12	71.61 $\pm$ 0.21	3.08 $\pm$ 0.65	3.77 $\pm$ 1.09

TABLE V

COMPARISON ON UTILITY (ACC AND AUC) AND FAIRNESS (Δ_{SP} AND Δ_{EO}) IN PERCENTAGE (%) WITH CREDIT AND AMINER-L. $\uparrow$ DENOTES THE LARGER, THE BETTER; $\downarrow$ DENOTES THE OPPOSITE. THE BEST RESULTS ARE RED AND BOLD-FACED. THE RUNNER-UPS ARE BLUE AND UNDERLINED.

Methods	Credit, Local				Credit, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{FL}	69.48 $\pm$ 0.93	68.71 $\pm$ 0.02	10.44 $\pm$ 0.06	9.62 $\pm$ 0.22	69.53 $\pm$ 0.92	69.66 $\pm$ 0.01	11.98 $\pm$ 0.12	10.62 $\pm$ 0.22
SAGE _{FL}	73.09 $\pm$ 0.19	72.98 $\pm$ 0.30	11.47 $\pm$ 0.87	10.06 $\pm$ 1.24	73.06 $\pm$ 0.18	73.68 $\pm$ 0.38	12.92 $\pm$ 1.50	10.88 $\pm$ 2.15
APNP _{FL}	69.52 $\pm$ 1.13	69.14 $\pm$ 0.04	12.49 $\pm$ 0.28	12.07 $\pm$ 0.60	69.57 $\pm$ 1.12	69.94 $\pm$ 0.05	16.00 $\pm$ 1.54	15.30 $\pm$ 2.35
GDC _{FL}	69.15 $\pm$ 0.75	68.47 $\pm$ 0.14	5.38 $\pm$ 0.44	5.25 $\pm$ 0.81	69.20 $\pm$ 0.75	69.46 $\pm$ 0.12	6.39 $\pm$ 1.03	5.94 $\pm$ 1.32
GAT _{FL}	72.38 $\pm$ 1.78	66.99 $\pm$ 1.66	5.13 $\pm$ 1.82	4.73 $\pm$ 2.45	69.04 $\pm$ 1.71	67.39 $\pm$ 3.41	7.91 $\pm$ 0.31	7.69 $\pm$ 1.78
NIFTY _{FL}	69.97 $\pm$ 2.33	61.01 $\pm$ 2.71	8.64 $\pm$ 2.61	8.64 $\pm$ 0.17	70.00 $\pm$ 1.06	60.83 $\pm$ 1.98	9.32 $\pm$ 1.69	9.11 $\pm$ 0.32
FairVGNN _{FL}	72.12 $\pm$ 0.45	68.46 $\pm$ 0.02	10.87 $\pm$ 0.78	10.07 $\pm$ 1.16	72.96 $\pm$ 2.27	69.23 $\pm$ 0.05	14.26 $\pm$ 2.38	13.10 $\pm$ 1.27
FairGNN _{FL}	76.66 $\pm$ 0.22	67.14 $\pm$ 0.74	6.34 $\pm$ 1.20	4.18 $\pm$ 1.53	76.68 $\pm$ 0.02	67.82 $\pm$ 0.45	7.02 $\pm$ 1.98	4.94 $\pm$ 1.18
FairSIN _{FL}	77.59 $\pm$ 0.81	67.84 $\pm$ 0.02	7.44 $\pm$ 1.08	5.04 $\pm$ 0.80	77.59 $\pm$ 0.79	68.39 $\pm$ 0.04	8.75 $\pm$ 1.68	6.25 $\pm$ 1.12
FED-PUB	66.30 $\pm$ 1.97	62.44 $\pm$ 2.16	7.30 $\pm$ 2.28	7.57 $\pm$ 1.59	66.35 $\pm$ 2.02	63.25 $\pm$ 1.64	8.38 $\pm$ 1.69	8.19 $\pm$ 1.75
FedGCN	69.48 $\pm$ 0.93	68.71 $\pm$ 0.02	10.44 $\pm$ 0.06	9.62 $\pm$ 0.22	69.53 $\pm$ 0.92	69.66 $\pm$ 0.01	11.98 $\pm$ 0.12	10.62 $\pm$ 0.22
FairFed	<u>77.82 $\pm$ 0.07</u>	64.66 $\pm$ 2.13	<u>2.84 $\pm$ 0.77</u>	<u>2.33 $\pm$ 1.11</u>	<u>77.79 $\pm$ 0.07</u>	64.82 $\pm$ 1.14	<u>2.78 $\pm$ 0.64</u>	<u>2.16 $\pm$ 1.05</u>
F ² GNN	76.74 $\pm$ 2.16	66.32 $\pm$ 2.08	6.85 $\pm$ 1.87	5.88 $\pm$ 1.47	76.75 $\pm$ 2.11	66.64 $\pm$ 0.47	7.86 $\pm$ 2.00	6.42 $\pm$ 1.16
F3GL	77.93 $\pm$ 0.01	<u>69.71 $\pm$ 2.14</u>	0.64 $\pm$ 0.27	0.35 $\pm$ 0.13	77.91 $\pm$ 0.01	<u>70.21 $\pm$ 1.54</u>	0.68 $\pm$ 0.31	0.34 $\pm$ 0.13

Methods	AMiner-L, Local				AMiner-L, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{FL}	90.89 $\pm$ 1.06	<u>92.60 $\pm$ 0.03</u>	12.11 $\pm$ 0.68	19.95 $\pm$ 0.02	90.88 $\pm$ 1.06	<u>93.77 $\pm$ 2.03</u>	6.19 $\pm$ 0.22	6.39 $\pm$ 2.27
SAGE _{FL}	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
APNP _{FL}	90.97 $\pm$ 2.11	92.46 $\pm$ 0.13	12.59 $\pm$ 0.79	19.23 $\pm$ 0.04	90.96 $\pm$ 1.11	93.64 $\pm$ 2.02	6.72 $\pm$ 0.47	6.50 $\pm$ 2.59
GDC _{FL}	87.83 $\pm$ 2.40	88.89 $\pm$ 2.86	9.02 $\pm$ 1.42	15.54 $\pm$ 1.92	87.80 $\pm$ 1.54	90.40 $\pm$ 1.78	6.92 $\pm$ 1.89	6.52 $\pm$ 1.97
GAT _{FL}	86.65 $\pm$ 0.31	83.34 $\pm$ 3.45	12.27 $\pm$ 0.74	20.20 $\pm$ 0.66	86.63 $\pm$ 0.30	85.03 $\pm$ 1.25	6.59 $\pm$ 0.32	4.59 $\pm$ 2.02
NIFTY _{FL}	87.54 $\pm$ 1.00	86.09 $\pm$ 2.01	5.86 $\pm$ 1.67	15.41 $\pm$ 1.19	87.52 $\pm$ 0.19	87.80 $\pm$ 1.00	4.66 $\pm$ 1.94	13.99 $\pm$ 1.36
FairVGNN _{FL}	89.34 $\pm$ 0.54	91.88 $\pm$ 2.09	13.18 $\pm$ 1.41	18.33 $\pm$ 1.15	89.43 $\pm$ 1.52	93.09 $\pm$ 1.80	7.66 $\pm$ 1.43	<u>4.24 $\pm$ 0.49</u>
FairGNN _{FL}	85.81 $\pm$ 1.75	89.03 $\pm$ 1.60	6.04 $\pm$ 0.48	<u>10.24 $\pm$ 1.55</u>	85.76 $\pm$ 1.25	90.47 $\pm$ 1.79	6.03 $\pm$ 0.02	6.11 $\pm$ 0.51
FairSIN _{FL}	84.62 $\pm$ 1.63	78.06 $\pm$ 1.15	<u>5.08 $\pm$ 1.70</u>	10.90 $\pm$ 1.95	84.58 $\pm$ 1.02	79.99 $\pm$ 1.75	<u>3.31 $\pm$ 1.35</u>	7.47 $\pm$ 1.03
FED-PUB	91.09 $\pm$ 0.37	92.50 $\pm$ 0.23	11.90 $\pm$ 0.94	19.63 $\pm$ 2.20	91.07 $\pm$ 1.37	93.75 $\pm$ 0.21	5.79 $\pm$ 0.25	7.04 $\pm$ 1.09
FedGCN	90.92 $\pm$ 0.60	92.64 $\pm$ 0.30	12.23 $\pm$ 0.52	20.12 $\pm$ 0.14	<u>90.90 $\pm$ 0.06</u>	93.80 $\pm$ 1.03	6.29 $\pm$ 0.19	6.33 $\pm$ 1.53
FairFed	87.28 $\pm$ 0.80	88.83 $\pm$ 2.84	11.27 $\pm$ 1.42	18.17 $\pm$ 2.21	87.26 $\pm$ 0.08	90.39 $\pm$ 1.52	6.04 $\pm$ 1.22	10.66 $\pm$ 1.76
F ² GNN	89.74 $\pm$ 0.34	90.48 $\pm$ 2.38	9.51 $\pm$ 1.89	21.16 $\pm$ 1.06	89.71 $\pm$ 0.34	91.80 $\pm$ 1.12	4.42 $\pm$ 1.66	11.69 $\pm$ 1.36
F3GL	<u>91.01 $\pm$ 1.31</u>	92.36 $\pm$ 0.58	4.02 $\pm$ 0.21	9.89 $\pm$ 1.09	89.99 $\pm$ 1.00	92.53 $\pm$ 0.86	2.11 $\pm$ 0.12	4.15 $\pm$ 1.15

C. Implementation Details and Parameter Settings

We implement all methods within a FedAvg framework. The implementation of all models utilizes PyTorch [57] and PyTorch-geometric [58]. The experiments are conducted on an Ubuntu operating system, with hardware specifications including an Intel(R) Core(TM) i7-14700 K and 2 RTX 4090 24 GB GPUs, and the CUDA version is 12.0. We used 200 global training rounds with the Adam optimizer [59], a learning rate of 0.0001, L2 regularization of 1×10^{-4} , and 10 local steps per round for federated settings. We selected 1,000 nodes for training and set the validation and testing sets to be 25% of the whole dataset. The original dataset was separated using the Louvain algorithm [60], and the clustering results with 10 clusters of similar sizes were used to obtain subgraphs to send to each client. The number of convolution layers was selected from {1, 2, 3}, and the number of neurons was selected from {16, 32, 64, 128}. We chose “region” and “gender” as sensitive features. This experimental setup ensures a rigorous and fair comparison between our proposed method and the baseline methods in federated learning environments.

D. Comparison Results

For assessing the utility of our method, we use node classification accuracy (ACC) and area under curve (AUC). For assessing fairness, we use Δ_{SP} and Δ_{EO} . Higher values in utility metrics

(ACC and AUC) indicate better performance, whereas lower values in fairness metrics (Δ_{SP} and Δ_{EO}) indicate better fairness.

1) *Fairness*: Tables IV–VI provide a comprehensive fairness evaluation of F3GL against baselines across two real-world datasets. F3GL consistently demonstrates outstanding local fairness and global fairness, with particularly notable improvements in Δ_{SP}^{Local} , Δ_{SP}^{Global} , Δ_{EO}^{Local} and Δ_{EO}^{Global} metrics across the evaluated datasets. These results highlight the robustness of F3GL in maintaining fairness during the learning process. The only exceptions are Δ_{EO}^{Global} in the Pokec-z-G dataset and Δ_{EO}^{Local} in the Pokec-n-G dataset, where F3GL shows runner-up performance. This slight variance does not overshadow the overall effectiveness of F3GL, as it still significantly outperforms the baselines in most scenarios. This comprehensive evaluation reaffirms F3GL’s ability to address fairness-related concerns within FedGNNs, ensuring both local and global fairness.

2) *Trade-Off Between Accuracy and Fairness*: To confirm our pursuit of a comprehensive assessment in both utility and fairness, our model achieves the best fairness while maintaining the highest accuracy and AUC among fairness-aware methods. Even when compared to graph convolution methods optimized solely for utility, our model consistently achieves runner-up results or comparable performance in most cases. As shown in Tables IV–VI, our proposed F3GL significantly enhances both fairness and utility. To provide a more visual representation of our effectiveness, we illustrate the details in Fig. 3. For

TABLE VI
COMPARISON ON UTILITY (ACC AND AUC) AND FAIRNESS (Δ_{SP} AND Δ_{EO}) IN PERCENTAGE (%) WITH POKEC-Z-R FOLLOWING TWO DIFFERENT FEDERATED FRAMEWORK. $\uparrow$ DENOTES THE LARGER, THE BETTER; $\downarrow$ DENOTES THE OPPOSITE. THE BEST RESULTS ARE RED AND BOLD-FACED. THE RUNNER-UPS ARE BLUE AND UNDERLINED.

Methods	Pokec-z-R, Local				Pokec-z-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+nova}	<u>70.83 $\pm$ 0.96</u>	73.60 $\pm$ 0.53	24.18 $\pm$ 3.32	18.79 $\pm$ 1.18	<u>68.46 $\pm$ 0.86</u>	74.32 $\pm$ 0.52	17.24 $\pm$ 1.95	10.37 $\pm$ 1.27
SAGE _{+nova}	71.28 $\pm$ 0.07	<u>74.01 $\pm$ 0.03</u>	24.18 $\pm$ 3.32	20.52 $\pm$ 3.00	69.20 $\pm$ 0.04	<u>75.11 $\pm$ 0.03</u>	14.56 $\pm$ 2.06	10.60 $\pm$ 1.06
APPNP _{+nova}	70.09 $\pm$ 0.87	74.16 $\pm$ 1.31	25.74 $\pm$ 0.20	22.75 $\pm$ 0.74	67.98 $\pm$ 0.70	75.14 $\pm$ 1.07	18.52 $\pm$ 1.01	10.39 $\pm$ 1.18
GDC _{+nova}	67.62 $\pm$ 1.79	72.69 $\pm$ 1.84	22.40 $\pm$ 2.24	21.94 $\pm$ 1.25	67.47 $\pm$ 2.35	73.56 $\pm$ 1.50	14.06 $\pm$ 1.65	9.75 $\pm$ 1.86
GAT _{+nova}	69.47 $\pm$ 1.01	71.72 $\pm$ 2.61	22.91 $\pm$ 2.09	21.02 $\pm$ 1.50	66.95 $\pm$ 1.73	72.58 $\pm$ 1.93	6.66 $\pm$ 1.57	6.00 $\pm$ 1.95
NIFTY _{+nova}	66.12 $\pm$ 2.15	68.71 $\pm$ 2.10	18.54 $\pm$ 1.64	16.35 $\pm$ 0.20	63.82 $\pm$ 2.22	70.57 $\pm$ 1.87	8.78 $\pm$ 1.56	6.45 $\pm$ 1.51
FairVGNN _{+nova}	66.48 $\pm$ 2.20	69.13 $\pm$ 0.24	31.01 $\pm$ 1.16	28.25 $\pm$ 0.03	63.86 $\pm$ 0.62	70.04 $\pm$ 0.12	26.08 $\pm$ 1.55	19.49 $\pm$ 1.20
FairGNN _{+nova}	68.94 $\pm$ 2.20	72.61 $\pm$ 0.76	22.32 $\pm$ 2.03	20.70 $\pm$ 1.07	68.72 $\pm$ 0.95	72.54 $\pm$ 0.95	15.25 $\pm$ 1.33	10.72 $\pm$ 1.31
FairSIN _{+nova}	67.98 $\pm$ 0.56	70.03 $\pm$ 0.63	23.61 $\pm$ 1.01	19.56 $\pm$ 1.59	65.72 $\pm$ 0.61	71.82 $\pm$ 0.29	13.97 $\pm$ 2.03	9.35 $\pm$ 1.87
FED-PUB	68.17 $\pm$ 2.73	72.00 $\pm$ 1.79	22.71 $\pm$ 2.40	20.77 $\pm$ 1.00	65.94 $\pm$ 2.75	73.30 $\pm$ 1.05	11.47 $\pm$ 1.41	7.77 $\pm$ 1.83
FedGCN	69.70 $\pm$ 1.37	73.06 $\pm$ 1.97	24.66 $\pm$ 1.18	21.97 $\pm$ 2.79	67.34 $\pm$ 0.54	72.88 $\pm$ 0.09	15.24 $\pm$ 2.14	9.72 $\pm$ 0.30
FairFed	59.44 $\pm$ 1.60	63.70 $\pm$ 2.13	20.12 $\pm$ 1.32	18.69 $\pm$ 1.69	65.48 $\pm$ 2.63	69.72 $\pm$ 0.70	2.92 $\pm$ 1.28	<u>3.08 $\pm$ 1.21</u>
F ² GNN	68.43 $\pm$ 0.99	72.74 $\pm$ 2.31	<u>16.50 $\pm$ 0.78</u>	<u>16.31 $\pm$ 0.28</u>	67.71 $\pm$ 0.19	73.40 $\pm$ 0.39	<u>2.66 $\pm$ 1.02</u>	4.49 $\pm$ 0.25
F3GL	69.45 $\pm$ 0.05	73.05 $\pm$ 0.68	15.62 $\pm$ 2.04	12.75 $\pm$ 2.98	67.69 $\pm$ 0.05	73.84 $\pm$ 0.12	4.71 $\pm$ 0.36	2.56 $\pm$ 0.05

Methods	Pokec-z-R, Local				Pokec-z-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	Δ_{SP} (%) $\downarrow$	Δ_{EO} (%) $\downarrow$
GCN _{+prox}	70.75 $\pm$ 0.96	73.59 $\pm$ 0.54	24.14 $\pm$ 0.48	18.84 $\pm$ 1.80	<u>68.38 $\pm$ 0.86</u>	74.31 $\pm$ 0.53	17.36 $\pm$ 1.40	10.40 $\pm$ 1.75
SAGE _{+prox}	71.21 $\pm$ 0.10	<u>74.01 $\pm$ 0.03</u>	24.20 $\pm$ 1.38	20.37 $\pm$ 1.44	69.12 $\pm$ 0.07	<u>75.10 $\pm$ 0.03</u>	14.57 $\pm$ 1.09	10.49 $\pm$ 1.49
APPNP _{+prox}	70.10 $\pm$ 0.85	74.16 $\pm$ 1.30	25.76 $\pm$ 0.16	22.74 $\pm$ 0.76	67.99 $\pm$ 0.68	75.13 $\pm$ 1.07	18.50 $\pm$ 0.96	10.36 $\pm$ 1.10
GDC _{+prox}	70.46 $\pm$ 1.62	72.63 $\pm$ 1.84	22.56 $\pm$ 1.62	21.73 $\pm$ 2.25	68.12 $\pm$ 1.35	73.50 $\pm$ 1.49	13.98 $\pm$ 1.65	9.66 $\pm$ 1.86
GAT _{+prox}	69.41 $\pm$ 2.02	71.73 $\pm$ 1.62	22.88 $\pm$ 3.13	20.99 $\pm$ 1.56	66.91 $\pm$ 1.73	73.59 $\pm$ 1.93	16.69 $\pm$ 2.54	13.68 $\pm$ 2.01
NIFTY _{+prox}	67.23 $\pm$ 0.05	69.83 $\pm$ 0.30	19.78 $\pm$ 1.94	17.39 $\pm$ 1.63	64.98 $\pm$ 0.13	71.35 $\pm$ 2.01	11.23 $\pm$ 1.05	6.90 $\pm$ 2.03
FairVGNN _{+prox}	67.44 $\pm$ 1.04	69.52 $\pm$ 1.74	31.12 $\pm$ 1.50	28.31 $\pm$ 0.94	63.85 $\pm$ 1.47	70.29 $\pm$ 0.73	26.06 $\pm$ 0.71	19.27 $\pm$ 1.40
FairGNN _{+prox}	<u>70.84 $\pm$ 1.34</u>	71.59 $\pm$ 0.76	23.03 $\pm$ 1.09	20.28 $\pm$ 1.57	66.62 $\pm$ 0.96	73.53 $\pm$ 0.39	15.41 $\pm$ 1.43	10.65 $\pm$ 2.75
FairSIN _{+prox}	68.29 $\pm$ 1.18	70.65 $\pm$ 0.99	22.55 $\pm$ 1.05	20.45 $\pm$ 0.80	66.17 $\pm$ 0.19	72.30 $\pm$ 0.31	3.69 $\pm$ 1.77	2.95 $\pm$ 0.81
FED-PUB	68.17 $\pm$ 2.73	72.00 $\pm$ 1.79	22.71 $\pm$ 2.40	20.77 $\pm$ 1.00	65.94 $\pm$ 2.75	73.30 $\pm$ 1.05	11.47 $\pm$ 1.41	7.77 $\pm$ 1.83
FedGCN	69.70 $\pm$ 1.37	73.06 $\pm$ 1.97	24.66 $\pm$ 1.18	21.97 $\pm$ 2.79	67.34 $\pm$ 0.54	72.88 $\pm$ 0.09	15.24 $\pm$ 2.14	9.72 $\pm$ 0.30
FairFed	59.44 $\pm$ 1.60	63.70 $\pm$ 2.13	20.12 $\pm$ 1.32	18.69 $\pm$ 1.69	65.48 $\pm$ 2.63	69.72 $\pm$ 0.70	2.92 $\pm$ 1.28	<u>3.08 $\pm$ 1.21</u>
F ² GNN	68.43 $\pm$ 0.99	72.74 $\pm$ 2.31	<u>16.50 $\pm$ 0.78</u>	<u>16.31 $\pm$ 0.28</u>	67.71 $\pm$ 0.19	73.40 $\pm$ 0.39	<u>2.66 $\pm$ 1.02</u>	4.49 $\pm$ 0.25
F3GL	68.83 $\pm$ 1.28	72.06 $\pm$ 0.67	14.25 $\pm$ 0.05	11.12 $\pm$ 2.01	67.38 $\pm$ 1.15	74.11 $\pm$ 0.54	0.18 $\pm$ 0.04	0.62 $\pm$ 0.06

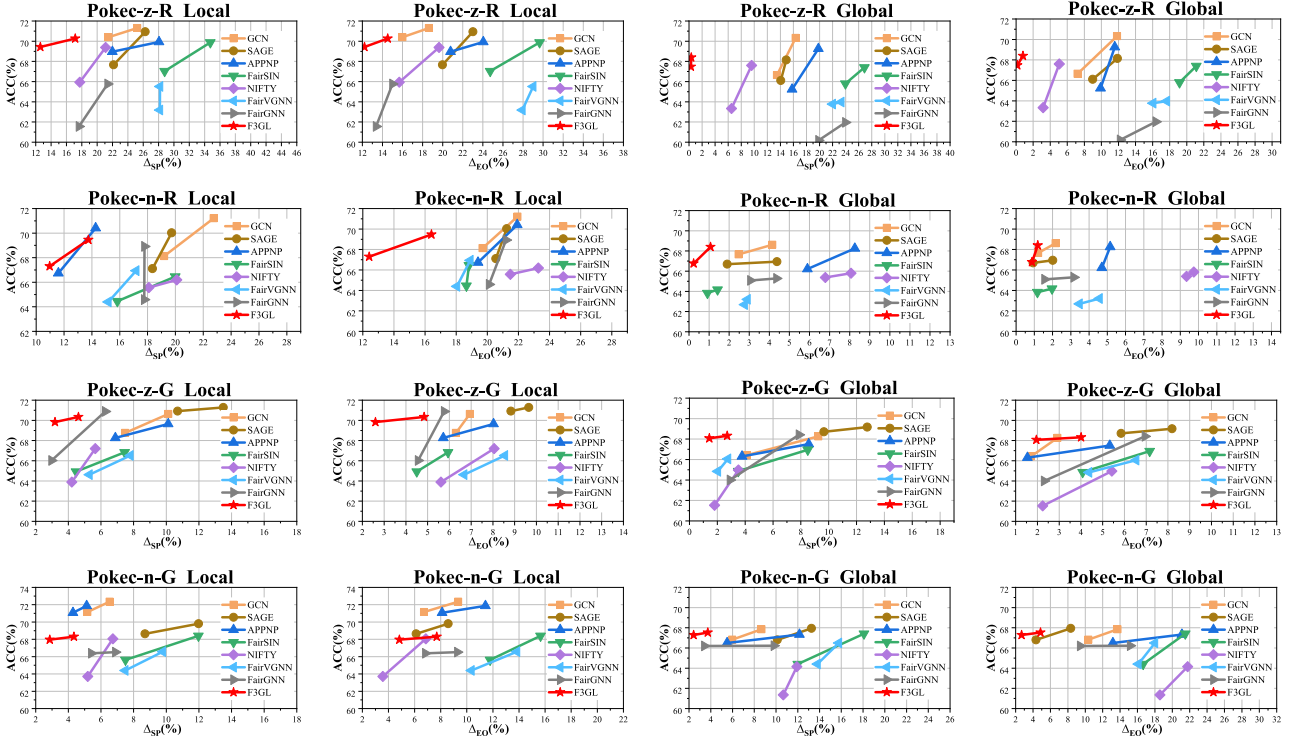


Fig. 3. The accuracy and fairness trade-off. The upper-left corner is preferable.

fairness evaluation, we employ metrics Δ_{SP}^{Local} , Δ_{SP}^{Global} , Δ_{EO}^{Global} , and Δ_{EO}^{Global} . Notably, the upper-left corner point in these graphs symbolizes the optimal scenario, characterized by the highest accuracy and highest prediction fairness. In the vast majority of cases, our results reside in the upper-left quadrant, demonstrating that our method successfully enhances both local and global fairness while minimizing the sacrifice in utility. This graphical representation underscores our method's capability to achieve a balanced trade-off between fairness and utility, ensuring robust performance across different evaluation metrics.

3) *Multi-Sensitive Features*: F3GL continues to deliver superior results even when handling multi-sensitive features. In our concurrent experiments, Table IV presents the results of four fairness assessment metrics for the sensitive features of gender and region, comparing our proposed F3GL method with various baseline models across two real-world datasets. F3GL consistently maintains excellent local fairness and global fairness. These results emphasize that F3GL not only improves local and global fairness during the learning process but also sustains a high level of fairness when multi-sensitive features are considered. This comprehensive evaluation demonstrates that F3GL effectively addresses fairness concerns related to multi-sensitive features in FedGNNs. This capability underscores F3GL's potential to provide fair and equitable solutions in federated graph learning scenarios.

The dataset **AMiner-L** involves sensitive attributes with multiple categorical classes rather than binary labels. While most traditional fairness metrics are designed for binary-sensitive attributes, these metrics can be naturally generalized to multi-class cases, as suggested in prior work [61]. The core idea behind this extension is to ensure fairness across all subgroup combinations by evaluating disparities among each pair of sensitive classes. One common strategy is to quantify fairness using the variance across all subgroups with respect to model predictions. Formally, in the multi-class sensitive feature setting, the adapted fairness metrics are defined as:

$$\begin{aligned}\Delta_{SP} &= \text{Var}_{i=1}^m \mathbb{P}(\hat{y} = 1 | s = i), \\ \Delta_{EO} &= \text{Var}_{i=1}^m \mathbb{P}(\hat{y} = 1 | y = 1, s = i),\end{aligned}\quad (26)$$

where m represents the total number of sensitive groups.

E. Ablation Study

F3GL, as an approach to harmonize the fairness and utility in FedGNNs, achieves its objectives through two key components: fairness-aware eigenvalue selection (F3GL_{FES}) and learnable Fourier features positional encoding (F3GL_{PE}). To assess the contributions of these two elements and the overall impact of F3GL, we conduct an ablation study. This study aims to determine whether both the F3GL_{FES} and F3GL_{PE} contribute to enhancing prediction fairness and utility. In the ablation study, we systematically remove each component independently to assess their individual impacts.

We examine the contribution of eigenvalue selection by evaluating F3GL without F3GL_{FES}, represented as **F3GL w/o FES**. The results in Table VII indicate that F3GL consistently outperforms F3GL w/o FES in terms of statistical parity (Δ_{SP}), equal

opportunity (Δ_{EO}), accuracy, and AUC. This demonstrates the necessity of F3GL_{FES} within F3GL. Interestingly, the performance improvement suggests that non-principal eigenvectors may introduce noise or irrelevant information, and discarding them not only enhances fairness but also improves overall utility. This finding supports our eigenvalue selection strategy as a beneficial trade-off mechanism between fairness and predictive performance. To assess the impact of F3GL_{PE}, we replace it with transformer position encoding (based on sine and cosine functions), denoted as **F3GL_{PE}**. The comparison in Table VII shows that F3GL outperforms F3GL_{PE} in terms of accuracy, AUC, Δ_{SP} and Δ_{EO} . This underscores the crucial role of F3GL_{PE} in enhancing the fairness and utility of GNN predictions. Thus, Fourier encoding yields better performance than alternative approaches such as Transformer-based encoding.

These results demonstrate the distinct contributions of each component to the overall performance of F3GL. When comparing F3GL to its ablated versions, F3GL consistently exhibits higher accuracy and AUC, and lower Δ_{SP} and Δ_{EO} . This observation highlights the integral role of both F3GL_{FES} and F3GL_{PE} in improving fairness and utility in F3GL.

F. Parameter Analysis

In this investigation, our primary aim is to scrutinize the impact of parameter e on our proposed F3GL, as shown in Table VIII. Specifically, we conduct a systematic parameter study focusing on the variable denoted as e across the two datasets. The parameter e assumes a pivotal role as it governs how original information from the adjacency matrix is retained. Moreover, it also plays a crucial role in representing the similarity between original sensitive features and ones after l layers of convolution. According to our theoretical analysis, as e increases, fairness initially hovers, followed by a subsequent decline. Experiment results verify our analysis. Notably, when non-principle eigenvalues are retained (i.e., larger e), fairness metrics degrade consistently, which empirically supports our assumption that non-principal eigenvectors tend to reintroduce correlations with sensitive attributes. This observation further validates our eigenvalue selection strategy, showing that retaining only the principle eigenvectors is beneficial for mitigating bias while preserving utility.

G. Training Cost Comparison

The whole eigendecomposition causes cubic complexity in terms of the number of nodes, resulting in a computational cost of $O(n^3)$. The eigenvector selection eliminates the need to rank the eigenvectors following the entire eigendecomposition process. The Arnoldi Package algorithm, which the fairness-aware eigenvalue selection of F3GL employs, achieves a time complexity of $O(ne^2)$ for eigendecomposition. F3GL only chooses principal eigenvectors, ensuring that e remains consistently below 10. We present the comparison results of whole eigendecomposition (denotes **WE**), $e = 5,000$, $e = 1,000$ and F3GL_{FES} in Table IX. According to the results, we observe that the time savings are more significant with larger datasets.

TABLE VII
COMPARISONS AMONG DIFFERENT COMPONENTS IN THE F3GL MODEL

Methods	Pokey-z-R, Local				Pokey-z-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	64.58 $\pm$ 1.07	67.49 $\pm$ 0.44	26.86 $\pm$ 2.78	24.55 $\pm$ 2.12	64.65 $\pm$ 1.48	68.14 $\pm$ 1.87	14.04 $\pm$ 2.84	14.07 $\pm$ 1.48
F3GL_{LPE}	69.49 $\pm$ 0.58	72.06 $\pm$ 0.69	21.17 $\pm$ 1.50	19.46 $\pm$ 2.95	67.84 $\pm$ 0.39	73.24 $\pm$ 0.64	1.73 $\pm$ 1.35	2.66 $\pm$ 1.80
F3GL	69.85 $\pm$ 0.42	73.62 $\pm$ 0.67	14.87 $\pm$ 2.27	13.37 $\pm$ 1.16	67.93 $\pm$ 0.46	73.68 $\pm$ 0.76	0.32 $\pm$ 0.02	0.47 $\pm$ 0.33
Methods	Pokey-z-G, Local				Pokey-z-G, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	64.45 $\pm$ 2.54	63.04 $\pm$ 1.45	20.34 $\pm$ 1.87	28.65 $\pm$ 0.94	63.08 $\pm$ 0.47	67.67 $\pm$ 1.50	6.35 $\pm$ 0.31	6.79 $\pm$ 1.82
F3GL_{LPE}	68.03 $\pm$ 0.30	68.10 $\pm$ 0.87	16.79 $\pm$ 1.59	24.47 $\pm$ 1.89	67.27 $\pm$ 0.25	71.14 $\pm$ 0.55	4.14 $\pm$ 1.95	5.39 $\pm$ 1.33
F3GL	68.38 $\pm$ 1.08	69.00 $\pm$ 2.05	12.36 $\pm$ 1.39	14.41 $\pm$ 2.00	67.59 $\pm$ 0.82	71.99 $\pm$ 1.31	0.65 $\pm$ 0.42	1.03 $\pm$ 0.16
Methods	Pokey-n-R, Local				Pokey-n-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	64.30 $\pm$ 2.85	68.11 $\pm$ 1.53	13.38 $\pm$ 0.25	10.19 $\pm$ 1.03	64.13 $\pm$ 1.75	70.03 $\pm$ 1.03	10.02 $\pm$ 1.10	10.31 $\pm$ 1.72
F3GL_{LPE}	67.94 $\pm$ 0.34	71.38 $\pm$ 0.09	15.40 $\pm$ 0.74	12.13 $\pm$ 0.79	66.45 $\pm$ 0.38	72.43 $\pm$ 0.22	13.53 $\pm$ 0.75	8.59 $\pm$ 0.67
F3GL	70.09 $\pm$ 0.25	73.13 $\pm$ 0.19	3.90 $\pm$ 0.72	3.72 $\pm$ 1.12	68.20 $\pm$ 0.12	74.54 $\pm$ 0.21	2.07 $\pm$ 0.64	2.98 $\pm$ 1.03
Methods	Pokey-n-G, Local				Pokey-n-G, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	63.04 $\pm$ 1.00	65.89 $\pm$ 0.63	13.50 $\pm$ 1.24	13.97 $\pm$ 1.62	63.13 $\pm$ 1.75	68.23 $\pm$ 0.98	10.83 $\pm$ 1.48	8.41 $\pm$ 1.81
F3GL_{LPE}	67.28 $\pm$ 0.36	68.99 $\pm$ 0.70	5.06 $\pm$ 1.61	6.92 $\pm$ 0.85	67.36 $\pm$ 0.28	71.22 $\pm$ 0.87	3.92 $\pm$ 1.13	8.86 $\pm$ 2.23
F3GL	68.13 $\pm$ 0.17	69.44 $\pm$ 0.16	3.60 $\pm$ 0.74	6.27 $\pm$ 1.43	67.41 $\pm$ 0.12	71.61 $\pm$ 0.21	3.08 $\pm$ 0.65	3.77 $\pm$ 1.09
Methods	Credit, Local				Credit, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	71.63 $\pm$ 0.57	64.12 $\pm$ 1.63	13.27 $\pm$ 1.74	12.62 $\pm$ 1.26	71.60 $\pm$ 1.57	64.77 $\pm$ 1.89	15.71 $\pm$ 0.84	13.75 $\pm$ 1.18
F3GL_{LPE}	75.61 $\pm$ 1.41	68.70 $\pm$ 1.70	3.83 $\pm$ 1.01	3.33 $\pm$ 0.58	76.59 $\pm$ 0.82	68.22 $\pm$ 0.59	3.93 $\pm$ 0.13	3.86 $\pm$ 0.23
F3GL	77.93 $\pm$ 0.01	69.71 $\pm$ 2.14	0.64 $\pm$ 0.27	0.35 $\pm$ 0.13	77.91 $\pm$ 0.01	70.21 $\pm$ 1.54	0.68 $\pm$ 0.31	0.34 $\pm$ 0.13
Methods	AMiner-L, Local				AMiner-L, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
F3GL w/o FES	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
F3GL_{LPE}	86.60 $\pm$ 1.14	86.82 $\pm$ 1.07	8.95 $\pm$ 0.08	17.68 $\pm$ 1.07	86.56 $\pm$ 1.28	88.56 $\pm$ 1.78	5.63 $\pm$ 0.31	6.70 $\pm$ 0.86
F3GL	88.01 $\pm$ 1.31	88.36 $\pm$ 0.58	4.02 $\pm$ 0.21	11.89 $\pm$ 1.09	87.99 $\pm$ 1.00	89.53 $\pm$ 0.86	2.11 $\pm$ 0.51	4.15 $\pm$ 1.25

TABLE VIII
THE ACCURACY, AUC, Δ_{SP} AND Δ_{EO} OF F3GL W.R.T. DIFFERENT PARAMETER e VALUES IN POKEC-Z-R.

Methods	Pokey-z-R, Local				Pokey-z-R, Global			
	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$	ACC(%) $\uparrow$	AUC(%) $\uparrow$	$\Delta_{SP}(\%)$ $\downarrow$	$\Delta_{EO}(\%)$ $\downarrow$
1	68.70 $\pm$ 1.28	72.98 $\pm$ 0.13	18.65 $\pm$ 2.31	13.47 $\pm$ 4.33	68.83 $\pm$ 1.53	73.22 $\pm$ 0.24	1.58 $\pm$ 1.22	1.33 $\pm$ 1.10
2	68.53 $\pm$ 0.78	72.73 $\pm$ 0.32	18.82 $\pm$ 1.88	17.37 $\pm$ 3.72	67.30 $\pm$ 0.97	72.69 $\pm$ 0.17	0.74 $\pm$ 1.19	1.70 $\pm$ 1.28
3	69.85 $\pm$ 0.42	73.62 $\pm$ 0.67	14.87 $\pm$ 2.27	13.37 $\pm$ 1.16	67.93 $\pm$ 0.46	73.68 $\pm$ 0.76	0.32 $\pm$ 0.02	0.47 $\pm$ 0.33
4	69.74 $\pm$ 0.92	73.62 $\pm$ 0.26	17.29 $\pm$ 1.90	15.56 $\pm$ 2.07	67.81 $\pm$ 0.97	73.74 $\pm$ 0.16	3.80 $\pm$ 2.46	3.59 $\pm$ 1.94
5	69.09 $\pm$ 0.39	73.57 $\pm$ 0.08	14.94 $\pm$ 3.56	12.78 $\pm$ 1.53	67.26 $\pm$ 0.34	73.77 $\pm$ 0.13	5.82 $\pm$ 1.31	3.45 $\pm$ 1.55
6	69.51 $\pm$ 0.17	73.72 $\pm$ 0.12	18.60 $\pm$ 2.30	17.11 $\pm$ 2.11	67.85 $\pm$ 0.33	74.25 $\pm$ 0.35	3.62 $\pm$ 1.34	2.63 $\pm$ 1.03
7	70.27 $\pm$ 0.39	73.88 $\pm$ 0.32	19.14 $\pm$ 2.50	14.53 $\pm$ 2.80	68.39 $\pm$ 0.42	74.44 $\pm$ 0.36	1.40 $\pm$ 2.59	0.96 $\pm$ 1.77
8	69.75 $\pm$ 0.60	73.50 $\pm$ 0.21	17.18 $\pm$ 2.06	14.76 $\pm$ 3.12	67.85 $\pm$ 0.69	73.82 $\pm$ 0.24	4.71 $\pm$ 2.38	4.19 $\pm$ 1.17
9	68.55 $\pm$ 0.69	73.35 $\pm$ 0.54	18.80 $\pm$ 2.36	17.07 $\pm$ 2.49	67.93 $\pm$ 0.67	73.71 $\pm$ 0.48	4.47 $\pm$ 1.57	4.53 $\pm$ 1.83
10	68.37 $\pm$ 0.80	72.42 $\pm$ 0.87	18.52 $\pm$ 1.05	17.16 $\pm$ 1.25	67.16 $\pm$ 0.73	72.92 $\pm$ 0.69	5.34 $\pm$ 1.52	5.77 $\pm$ 2.51
100	67.09 $\pm$ 0.44	70.73 $\pm$ 0.04	20.36 $\pm$ 1.48	20.53 $\pm$ 2.26	65.66 $\pm$ 0.27	71.05 $\pm$ 0.02	7.29 $\pm$ 2.53	7.08 $\pm$ 1.89
500	66.58 $\pm$ 0.38	69.98 $\pm$ 0.18	22.17 $\pm$ 1.08	21.43 $\pm$ 1.62	65.62 $\pm$ 0.23	70.92 $\pm$ 0.05	9.96 $\pm$ 1.48	8.82 $\pm$ 0.83
1,000	66.24 $\pm$ 0.24	69.88 $\pm$ 0.07	24.60 $\pm$ 1.55	21.85 $\pm$ 2.19	65.34 $\pm$ 0.31	70.14 $\pm$ 0.05	11.00 $\pm$ 0.68	11.37 $\pm$ 0.99
2,000	66.00 $\pm$ 0.31	68.49 $\pm$ 0.11	25.61 $\pm$ 0.94	21.01 $\pm$ 1.53	65.11 $\pm$ 0.37	69.90 $\pm$ 0.31	11.59 $\pm$ 2.15	12.01 $\pm$ 2.09
5,000	65.02 $\pm$ 1.37	67.77 $\pm$ 2.63	26.36 $\pm$ 0.90	24.22 $\pm$ 1.03	64.93 $\pm$ 1.22	68.83 $\pm$ 1.98	13.61 $\pm$ 1.70	12.55 $\pm$ 1.81

TABLE IX
RUNTIME

Methods	WE	$e = 5,000$	$e = 1,000$	F3GL _{FES}
Credit	431.31 s	3,227.79 s	304.86 s	0.45 s
Pokec-z	121.83 s	9,352.48 s	721.14 s	1.45 s
Pokec-n	109.47 s	10,608.11 s	689.45 s	1.42 s
AMiner-L	418.45 s	31,026.94 s	1,640.26 s	1.50 s

TABLE X
MEMORY

Methods	WE	$e = 5,000$	$e = 1,000$	F3GL _{FES}
Credit	3.35 GB	572.23 MB	114.45 MB	2.29 MB
Pokec-z	17.12 GB	1.26 GB	258.63 MB	5.17 MB
Pokec-n	16.51 GB	1.24 GB	253.95 MB	5.08 MB
AMiner-L	62.69 GB	2.42 GB	494.87 MB	9.90 MB

As shown in Table X, we further compare the size of pre-processed files generated under different decomposition strategies. It is evident that F3GL, through its eigenvalue selection mechanism, substantially reduces storage requirements. This

highlights its strong potential for deployment on large-scale graphs, where memory constraints are a common bottleneck. In particular, for datasets where using the full eigendecomposition would otherwise exceed available memory during inference, our

method remains fully functional. The comparison results clearly demonstrate that F3GL_{FES} achieves notable storage savings compared to both full-spectrum and partial-spectrum methods, establishing it as a practical and scalable solution for large-graph scenarios.

VII. CONCLUSION

In this work, we elucidate the disparity between original sensitive features and those after l layers of convolution operations, establishing a strong correlation with the largest magnitude eigenvalue of the adjacency matrix. Drawing inspiration from this foundational insight, we present F3GL, an innovative approach dedicated to enhancing both local fairness and global fairness without sacrificing utility within a federated framework. F3GL is built on a novel eigenvalue selection mechanism for sensitive features, which forms the core of fairness-aware eigenvalue selection. These meticulously crafted elements synergistically elevate the performance of FedGNNs, positioning F3GL as a potent and robust solution compared to contemporary baselines. In future work, we plan to further refine the efficiency of the F3GL approach and extend its applicability to situations with limited sensitive features. By optimizing the algorithm for even more efficient computation and expanding its scope to handle diverse scenarios, we aim to make F3GL an even more robust and versatile tool for achieving fairness in federated graph learning.

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